SHAREMATE: A SOCIALLY CONNECTED SMART MOBILITY AND COMMUNITY SUPPORT APP

ACADEMIC NOTE INTELLIGENCE ENGINE (ANIE)

INTELLIGENT ACADEMIC NOTE VALIDATION, DIFFICULTY CLASSIFICATION AND PERSONALISED RECOMMENDATION

INDIVIDUAL THESIS

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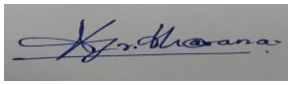
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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis represents my own research contribution, properly acknowledges all sources used, and discloses the use of automated tools in accordance with institutional policy.

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ACADEMIC INTEGRITY

This thesis was prepared with reference to the ShareMate project paper and the implemented ANIE component. All screenshots, diagrams, tables and references included in this submission should correspond to verified project evidence and institutional formatting requirements.

ABSTRACT

University students regularly exchange lecture notes, summaries and tutorial documents using fragmented channels in which quality, completeness and suitability for different learners are difficult to establish. ShareMate is a university-focused community platform that combines transportation, shared rides, academic resources and collaborative shopping. This individual thesis focuses on its Academic Note Intelligence Engine (ANIE), the note-sharing component designed to validate student-contributed study materials, assign difficulty levels, recommend relevant resources and encourage high-quality contributions. The proposed workflow accepts PDF and image-based notes, extracts readable text, compares content with authorised lecturer-provided materials, evaluates conceptual correctness, structure, reference validity, completeness and readability, and produces a weighted quality score. Resources are grouped into Beginner, Intermediate and Advanced levels. Topic-based recommendations, guided learning paths and a token-based contribution mechanism support discovery and sustained participation. The component is designed as an independently accessible service within ShareMate's API-oriented microservice architecture. The source project paper reports an ANIE classification accuracy of 89.6% and recommendation accuracy of 87.8%; these are project-level reported figures and require independent replication and experimental documentation in the final individual thesis. This draft specifies reproducible evaluation procedures, security and ethical safeguards, integration contracts, implementation requirements and screenshot locations. Its expected contribution is a more trustworthy and level-appropriate academic resource-sharing experience within a larger student community platform.

Keywords: ShareMate, Academic Note Intelligence Engine, ANIE, academic note sharing, content validation, difficulty classification, recommender systems, education technology, explainable scoring.

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1 INTRODUCTION

1.1 Overall project context

University students often face challenges in managing their daily transportation, accessing academic materials, and meeting other essential needs while keeping their expenses affordable. Although students regularly share resources and help one another through informal groups and online platforms, these activities are usually managed through separate applications. This can make it difficult for students to find reliable information, access useful resources, and connect with others within their university community.

To address these challenges, the ShareMate project introduces a university-focused platform that brings several resource-sharing services together in one application. The system consists of four main components: the FriendZone LiveTransit Share Engine (FLTS), the Academic Note Intelligence Engine (ANIE), FriendZone Ride, and a collaborative grocery shopping service. Each component is designed to support a different aspect of student life while encouraging collaboration and resource sharing within the university community [S1].

ShareMate follows a modular architecture in which individual services communicate through Application Programming Interfaces (APIs). This allows each component to be developed, maintained, and improved independently without affecting the overall system. Through a shared mobile platform, ShareMate aims to provide students with a more convenient way to access academic resources, share transportation, manage everyday needs, and participate in a trusted university community [S1].

1.2 Individual component context

This thesis examines the Academic Note Intelligence Engine (ANIE), the part of ShareMate devoted to sharing and assessing academic notes. University students often share lecture notes, study summaries, tutorials, and other learning materials to support one another in their studies. However, finding useful and reliable notes can be challenging, especially when students have limited information about the quality or difficulty of the shared materials.

The ANIE component aims to address this challenge by providing students with a more organised and reliable way to share and access academic resources. Rather than functioning only as a platform for uploading and downloading documents, it is designed to evaluate the quality of student-uploaded notes by comparing their content with lecturer-provided learning materials. This process considers factors such as content accuracy, relevance, completeness, clarity, and readability.

In addition, ANIE classifies academic notes into three difficulty levels: Beginner, Intermediate, and Advanced. This allows students to select learning materials according to their understanding of a particular subject. The component also provides topic-based recommendations to help students discover related resources and introduces a reward mechanism to encourage students to contribute useful, high-quality notes.

By bringing these features together, ANIE aims to create a more supportive academic environment where students can share knowledge, find suitable learning materials, and make better decisions about the resources they use for their studies [S1].

1.3 Problem statement

Sharing academic notes has become a common practice among university students, particularly through messaging applications, social media groups, and online storage platforms. Although these methods make it easy to exchange learning materials, they do not always help students determine whether the shared notes are accurate, complete, relevant, or suitable for their level of understanding.

One of the main challenges is the lack of a reliable method for evaluating the quality of student-uploaded academic materials. Notes shared by students may contain missing information, unclear explanations, or content that does not fully match the topics covered in lectures. Since most informal sharing platforms do not provide an automated content validation process, students often need to review several documents before finding material they can confidently use.

Another challenge is that students have different levels of knowledge and learning needs. A resource that is useful for an advanced learner may be difficult for a

beginner to understand. However, conventional note-sharing methods generally do not classify materials according to their difficulty or provide personalised recommendations based on learning requirements. As a result, students may spend unnecessary time searching for appropriate study resources.

Furthermore, many existing sharing methods provide limited encouragement for students to contribute well-organised and high-quality academic content. Without a structured evaluation and reward mechanism, it can be difficult to identify valuable contributions and motivate continued knowledge sharing.

Therefore, this research focuses on developing an intelligent academic note-sharing component that can assess uploaded notes against authorised lecturer-provided materials, identify their quality and relevance, classify them according to difficulty, and recommend suitable learning resources. Through the Academic Note Intelligence Engine (ANIE), the proposed solution aims to improve the reliability, accessibility, and usefulness of academic note sharing within the ShareMate university community.

1.4 Research gap

Academic note sharing and digital learning resource management have become important parts of university education. Existing platforms allow students to upload, store, and access educational materials, while previous research has explored artificial intelligence, recommendation systems, and personalised learning approaches to improve access to academic resources [R1]–[R4]. However, bringing these capabilities together in a way that supports reliable and meaningful academic note sharing remains an important area for further development.

One of the key challenges is ensuring the quality and reliability of student-uploaded notes. Although digital repositories make learning materials accessible, storing and sharing documents alone does not guarantee that their content is accurate, complete, or consistent with lecturer-provided materials. Students may therefore find it difficult to determine which notes are suitable for their studies without manually reviewing the content.

Another area that requires attention is the relationship between academic content quality and individual learning needs. Previous research has explored personalised educational recommendations, but the ShareMate project identifies an opportunity to combine note-quality assessment with difficulty classification and topic-based recommendations in a single workflow. This would allow students to consider not only whether a resource is relevant to a subject, but also whether it matches their current level of understanding.

Encouraging students to contribute useful academic materials is another important consideration. A structured recognition and reward mechanism can help promote active participation and motivate students to share well-organised learning resources. Integrating such incentives with automated note assessment creates an opportunity to support both resource quality and continued student engagement.

Based on these considerations, the research gap addressed by this study is the need for an integrated, university-focused academic note-sharing framework that combines automated content validation, lecturer-material comparison, note-quality

scoring, difficulty classification, personalised topic-based recommendations, and contributor incentives.

To address this gap, the proposed Academic Note Intelligence Engine (ANIE) brings these functions together within the ShareMate platform. Rather than focusing only on document storage or recommendation, ANIE aims to provide a more structured approach to evaluating, organising, and discovering academic notes while encouraging students to contribute valuable learning materials [S1].

The identified gap primarily concerns the integration of these features within the proposed university-sharing environment. It does not suggest that individual features are entirely absent from existing educational systems. Further comparison with related research and existing applications is necessary to establish the specific contribution and novelty of the proposed approach

1.5 Aim and objectives

Research Aim

The main aim of this research is to design, develop, and evaluate an intelligent academic note-sharing component that improves the quality, reliability, and accessibility of educational resources shared among university students. Through the Academic Note Intelligence Engine (ANIE), this study aims to help students identify trustworthy learning materials, discover notes that match their knowledge levels, and access relevant academic content more efficiently within the ShareMate platform.

Specific Objectives

To achieve this aim, the following research objectives have been identified:

- **O1 – Academic Note Uploading and Management:** To develop a secure and user-friendly mechanism that allows students to upload, organise, store, and access academic notes in PDF and image formats.
- **O2 – Automated Note Quality Assessment:** To develop a method for extracting text from uploaded documents and evaluating their accuracy, relevance,

completeness, and clarity by comparing the content with authorised lecturer-provided learning materials.

- **O3 – Difficulty Level Classification:** To design and implement a classification mechanism that categorises academic notes into three difficulty levels: Beginner, Intermediate, and Advanced, helping students select resources suitable for their current understanding.
- **O4 – Personalised Note Recommendation:** To develop a topic-based recommendation mechanism that helps students discover relevant academic notes and follow suitable learning sequences based on their learning interests and requirements.
- **O5 – Quality-Based Contribution and Reward System:** To introduce a rating and token-based incentive mechanism that recognises valuable academic contributions and encourages students to share useful, well-organised learning materials.
- **O6 – System Integration and API Development:** To develop and document the necessary Application Programming Interfaces (APIs) to connect the ANIE component with the overall ShareMate application, allowing students to access academic resources through a common platform.
- **O7 – System Evaluation and Validation:** To evaluate the effectiveness of the proposed component through reproducible testing, focusing on note-quality assessment agreement with human judgement, difficulty classification accuracy, recommendation relevance, system response time, and user satisfaction.

1.6 Research questions

This research aims to address the challenges associated with the reliability, organisation, and accessibility of student-shared academic resources. To guide the development and evaluation of the Academic Note Intelligence Engine, the following research questions have been formulated:

RQ1 – Academic Note Quality Assessment

How effectively can the proposed ANIE component evaluate the quality and reliability of student-uploaded academic notes when compared with assessments made by lecturers or human evaluators using predefined academic criteria?

RQ2 – Difficulty Level Classification

How accurately can academic notes be classified into Beginner, Intermediate, and Advanced difficulty levels based on their content and complexity?

RQ3 – Personalised Academic Resource Recommendation

To what extent can topic-based recommendations that consider academic content quality improve the relevance of suggested notes compared with traditional keyword-based search methods?

RQ4 – System Performance and Integration

How effectively can the Academic Note Intelligence Engine be integrated into the ShareMate platform while maintaining acceptable levels of usability, security, and system response time?

1.7 Scope, assumptions and limitations

This research focuses on the development of the Academic Note Intelligence Engine (ANIE), which is designed to improve the way university students share, evaluate, and access academic learning materials within the ShareMate platform. The scope of this study covers the main functions required to manage student-uploaded notes and support students in finding reliable and suitable resources for their studies.

The component is designed to support academic materials such as lecture notes, study summaries, tutorials, and other educational documents uploaded in PDF or image formats. It includes the collection of relevant document information, text extraction, and the comparison of uploaded content with authorised lecturer-provided learning materials. Based on this process, ANIE aims to assess note quality by considering factors such as content relevance, completeness, clarity, readability, and consistency with the reference materials.

Another important part of the research is the classification of academic notes according to their difficulty levels. The system is designed to categorise resources as Beginner, Intermediate, or Advanced, allowing students to identify materials that match their current knowledge and learning needs. Topic-based recommendations are also included to help students discover related academic resources more conveniently.

In addition, the research covers a rating and token-based reward mechanism that encourages students to contribute useful and well-organised notes. The development of APIs for communication with other ShareMate services, together with functional testing and performance evaluation, also forms part of the proposed scope.

However, this research has several limitations. The automated quality assessment process is intended to provide guidance to students rather than act as an official academic certification. Although uploaded notes can be compared with lecturer-provided materials, the system cannot guarantee that every statement in a document is factually correct or that all relevant academic content has been covered.

The effectiveness of the component also depends on the quality and availability of reference materials and uploaded notes. Documents containing unclear images, poor formatting, handwritten content, or incomplete information may affect the accuracy of text extraction and subsequent assessment. Similarly, the performance of difficulty classification and recommendations may vary depending on the range of academic subjects and learning materials used for evaluation.

The research does not include automatic copyright clearance for uploaded documents, formal grading of students, or decision-making in medical and other high-risk areas. Responsibility for providing authorised learning materials and respecting copyright requirements remains an important consideration.

The specific machine learning architecture, dataset size, training and testing arrangements, and deployment configuration must be confirmed through the final implementation and experimental records. Therefore, the proposed functionality described in this thesis should be distinguished from features that have been fully implemented and experimentally validated.

Within these boundaries, this research aims to provide a practical and structured approach to academic note sharing that supports students in making more informed decisions about the learning materials they use.

1.8 Individual contribution

My individual contribution to the ShareMate research project focuses on the design and development of the Academic Note Intelligence Engine (ANIE), which is responsible for supporting intelligent academic note sharing within the university community. While the overall ShareMate platform consists of several interconnected services, this research specifically addresses the challenges of evaluating, organising, and recommending student-shared academic materials.

A major part of my contribution involves designing a document management and assessment process that allows students to upload academic notes and receive useful information about their quality. This includes the proposed workflow for extracting text from PDF and image-based documents, comparing uploaded content with authorised lecturer-provided materials, and assessing factors such as relevance, completeness, clarity, and readability.

Another important contribution is the development of a difficulty classification approach that categorises academic notes into Beginner, Intermediate, and Advanced levels. This feature is intended to help students select materials based on their existing knowledge and understanding rather than relying only on document titles or descriptions.

The research also includes the design of a topic-based recommendation mechanism that helps students discover related notes and identify suitable learning resources. Alongside this, a quality-aware rating and token-based incentive mechanism is proposed to recognise valuable contributions and encourage students to share useful academic materials.

My contribution further includes designing an API interface through which the ANIE component can communicate with the main ShareMate application. This supports the

integration of note uploading, automated assessment, difficulty classification, recommendations, and contributor feedback into the wider platform.

The individual research also covers the planned testing and evaluation of these functions, including the accuracy of difficulty classification, the relevance of recommended notes, the agreement between automated quality assessments and human evaluations, and the overall usability and performance of the component.

The remaining ShareMate services, including the FriendZone LiveTransit Share Engine, FriendZone Ride, and collaborative grocery shopping module, are developed as separate responsibilities within the research team. My work concentrates on the academic resource-sharing component while supporting its integration with the overall ShareMate system.

Through this individual contribution, I aim to demonstrate how intelligent document assessment, difficulty classification, and personalised recommendations can be combined to make academic note sharing more reliable, accessible, and useful for university students.

1.9 Thesis structure

This thesis is organised into nine chapters, each presenting an important stage of the research and development of the Academic Note Intelligence Engine (ANIE) within the ShareMate platform.

Chapter 1 – Introduction provides an overview of the ShareMate project and introduces the ANIE component. It explains the research problem, identifies the research gap, outlines the aim and objectives, and describes the scope, limitations, and individual contribution of this study.

Chapter 2 – Literature Review and Related Work examines previous research and existing technologies related to academic note sharing, automated content validation, educational resource recommendation, difficulty classification, and incentive-based learning platforms. It also highlights the limitations of existing approaches and explains how they relate to the research.

Chapter 3 – Research Methodology describes the research approach and methods used to design and evaluate ANIE. It covers data collection, document preprocessing, note-quality assessment, difficulty classification, recommendation methods, and the evaluation procedures considered for the component.

Chapter 4 – Product and Solution Design presents the overall system architecture, functional and non-functional requirements, component design, and the interaction between ANIE and the other ShareMate services.

Chapter 5 – Implementation, Integration and Testing explains the development of the ANIE component, including its main functions, technologies, API integration, and software testing procedures. It also presents relevant system interfaces and implementation evidence.

Chapter 6 – Results and Validation presents the available research findings and evaluates the performance of the academic note-sharing component. It distinguishes the results reported in the overall ShareMate research paper from independently verified individual component results and identifies any evaluations for which sufficient evidence is not yet available.

Chapter 7 – Discussion interprets the research findings, considers their relationship to the research objectives and previous studies, and discusses the strengths, limitations, and practical implications of the developed approach.

Chapter 8 – Commercialisation and Sustainability explores the potential application of ANIE in university learning environments, its value to students and educational institutions, and considerations related to long-term sustainability, scalability, and responsible use.

Chapter 9 – Conclusion and Future Directions summarises the research, reviews the achievement of the objectives, highlights the individual contribution of ANIE to the ShareMate platform, and identifies opportunities for future improvements.

The thesis concludes with references, a glossary, and appendices containing supporting technical documentation and relevant research materials.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review examines previous research related to academic note sharing, automated document quality assessment, difficulty classification, and personalised learning resource recommendation. The review considers what existing approaches do well, where they fall short, and what ANIE can learn from them.

The review is primarily based on the ShareMate research paper and the relevant studies referenced within it [S1], [R1]–[R4]. Particular attention is given to educational recommendation systems, content evaluation methods, and the use of intelligent technologies to organise academic resources.

The studies are discussed in relation to ANIE, with attention to both their useful methods and their limitations. Together, they inform ANIE's approach to note validation, quality assessment, difficulty classification, topic-based recommendations and contributor incentives.

2.2 Domain and theoretical foundations

A useful academic note needs more than a clear title. Accuracy, coverage of the topic, organisation, readability and dependable references all matter. Looking at these qualities separately helps students judge whether a note will support their learning. Although an overall quality score can make it easier to compare notes, individual assessment criteria should also be considered to provide a clearer understanding of their strengths and limitations.

Automated document assessment involves extracting text from uploaded files, processing the content, identifying important topics, and comparing the information with relevant reference materials. For scanned PDFs and images, Optical Character Recognition (OCR) is used to extract text. However, errors during text extraction can affect the reliability of the assessment, making document quality checks an important part of the process.

Recommendation systems help students discover suitable learning materials by considering content relevance and individual learning needs. Content-based

recommendations identify resources with similar topics, while collaborative approaches use previous user interactions to suggest materials. Combining these methods can improve recommendations when sufficient data is available. In an academic environment, recommendations should also consider course relevance, topic coverage, and difficulty level rather than relying only on popularity

2.3 Related research

Several studies have explored how recommendation systems can help users discover relevant information and learning resources. Adomavicius and Tuzhilin [R1] reviewed content-based, collaborative, and hybrid recommendation approaches, providing a foundation for understanding how academic notes can be recommended based on their content and user preferences.

Zhai et al. [R2] explored personalised learning resource recommendations based on students' knowledge levels, while Wei et al. [R3] examined the use of artificial intelligence to support personalised online learning. Similarly, Liang et al. [R4] discussed recommendation approaches designed for higher education. These studies highlight the importance of considering students' learning needs when suggesting educational resources.

Although these studies provide useful foundations for personalised recommendations, the ShareMate project focuses on bringing several academic note-sharing functions together within a single platform. The Academic Note Intelligence Engine (ANIE) combines note-quality assessment, difficulty classification, and topic-based recommendations to help students find suitable learning materials [S1]. However, the effectiveness of these functions must be evaluated through appropriate testing, including comparison with human assessments and examination of recommendation relevance.

2.3.1 Traditional Academic Note Sharing and Repository Systems

Traditional academic repositories and shared drives make study materials easier to distribute, but access alone does not establish whether a student-authored note is accurate or suitable for a particular learner. This distinction is important for ANIE: the aim is not to replace familiar file-sharing tools, but to supplement discovery with information about note quality and subject relevance.

2.3.2 Machine Learning and AI-Based Academic Note Validation

Automated document analysis can support text extraction, content comparison and the organisation of large collections of student notes. For ANIE, comparison with lecturer-authorised resources provides useful reference context; however, a similarity score is not the same as proof of factual correctness. Quality feedback therefore needs clear assessment criteria and an explanation of what the system could or could not check [S1].

2.3.3 Educational Content Difficulty Classification

Students studying the same topic may need resources at different levels of detail. ANIE uses Beginner, Intermediate and Advanced categories to make this distinction visible during note selection [S1]. A useful classification process should reflect the complexity of terminology, prerequisite knowledge and explanation, and its reliability should be evaluated against consistent labels.

2.3.4 Personalised Academic Resource Recommendation Systems

Adomavicius and Tuzhilin describe content-based, collaborative and hybrid recommendation strategies [R1]. Educational studies cited in the ShareMate paper also emphasise learner knowledge and context when suggesting resources [R2]–[R4]. ANIE connects these ideas to course- and topic-based note discovery, with quality and difficulty information intended to support more informed selection [S1].

2.3.5 Trust, Reputation and Incentive-Based Academic Sharing

Students may be more willing to contribute resources when useful work is recognised. ShareMate describes top-rated academic content and token-based incentives as ways to encourage participation [S1]. These features also create a need

for fair rating rules and safeguards against duplicate, misleading or unattributed uploads; rewards should not be mistaken for independent academic approval.

2.4 Existing systems, products and techniques

Existing platforms such as Google Drive and other cloud storage services allow students to upload, store, and share academic materials easily. Similarly, Learning Management Systems (LMS) help universities organise lecture materials and provide access to educational resources. However, these platforms may not provide dedicated features for automatically evaluating student-uploaded notes, identifying their difficulty levels, or recommending materials based on individual learning needs.

AI-based tools and recommendation systems can also help students discover and organise educational content. However, their recommendations do not necessarily consider the accuracy of shared notes or compare them with authorised lecturer-provided materials. This highlights the importance of combining content evaluation with resource recommendation in an academic environment.

The Academic Note Intelligence Engine (ANIE) within ShareMate is designed to address these needs by bringing together note-quality assessment, lecturer-material comparison, difficulty classification, topic-based recommendations, and contribution rewards. These features aim to provide students with a more organised and reliable academic note-sharing experience [S1].

Existing System / Approach	Main Strength	Limitation in Relation to ANIE
Cloud File Sharing (e.g., Google Drive)	Easy uploading, storage, and sharing of academic documents	Does not inherently evaluate academic note quality or difficulty levels
Learning Management Systems (LMS)	Provides organised access to course materials	Peer contributions and personalised note recommendations may be limited
Keyword-Based Search	Enables quick document retrieval using keywords	May not consider note quality, difficulty level, or individual learning needs
Generic Recommendation Systems	Suggests relevant resources based on content and user preferences	May not validate uploaded notes against lecturer-provided materials
Proposed ANIE [S1]	Combines note validation, quality scoring, difficulty classification, topic-based recommendations, and contributor rewards	Depends on reliable lecturer materials and accurate automated assessments

Table 2.2: Comparison of Existing Systems and the Proposed ANIE Component

Table 2.2 highlights the strengths and limitations of existing academic resource-sharing approaches. Unlike systems that focus mainly on storage or recommendations, the proposed ANIE component aims to combine note validation, difficulty classification, and personalised recommendations within a single university-focused platform.

2.5 Gap synthesis and conceptual framework

The literature review highlights the importance of reliable academic content, personalised learning, and effective resource sharing in higher education. Although existing systems support document storage, content discovery, and learning resource recommendations, there is an opportunity to combine these functions with automated note-quality assessment and difficulty classification within a single university-focused platform.

To address this gap, the Academic Note Intelligence Engine (ANIE) brings together lecturer-provided reference materials, student-uploaded notes, and course information. The system is designed to extract and analyse document content, evaluate note quality, classify difficulty levels, and recommend relevant academic resources. Student ratings and feedback can further support resource improvement, while contribution rewards encourage the sharing of useful learning materials.

The main contribution of ANIE is the integration of these features into a structured academic note-sharing workflow within ShareMate. Rather than introducing each technique as an entirely new concept, the research focuses on combining content validation, difficulty classification, recommendations, and student participation to support a more reliable and accessible learning environment.

3 RESEARCH METHODOLOGY

3.1 Research design

This research followed a design-science approach to develop and evaluate the Academic Note Intelligence Engine (ANIE) within the ShareMate platform. The study focused on improving the reliability and accessibility of academic notes shared among university students through automated note-quality assessment, difficulty classification, and learning resource recommendations.

The research involved developing the ANIE component and testing its functionality through the ShareMate mobile application. Practical demonstrations were carried out to examine academic note uploading, automated document processing, quality score generation, subject-based note browsing, token-based access, and document viewing. These tests provided evidence of the component's operation within the application.

The evaluation also considered the classification and recommendation performance reported in the ShareMate research paper. The implementation and functional testing are discussed in Chapter 5, while the available performance results and their limitations are presented in Chapter 6.

3.2 Data, participants and study materials

This research used academic notes and educational materials to develop and test the Academic Note Intelligence Engine (ANIE) within the ShareMate platform. The study focused on student-uploaded notes, including PDF documents and image-based learning materials, along with course information and relevant academic reference resources.

The uploaded materials were used to test document processing, automated note-quality assessment, difficulty classification, and topic-based recommendations. The evaluation focused on how effectively the system could analyse academic content, generate quality scores, and help students discover suitable learning resources.

The system also considered information such as note identifiers, course details, document formats, quality scores, difficulty levels, and topic information. These details supported the organisation and management of academic resources within

ShareMate. Attention was also given to protecting student information and maintaining the reliability of shared learning materials.

Table 3.1: Academic Note Data and Metadata in ANIE

Data Field	Purpose
Note ID	Identifies each uploaded academic note
Course / Subject	Organises notes according to academic subjects
File Type	Identifies PDF or image-based documents
Extracted Text	Supports academic content analysis
Quality Score	Represents the automated note assessment
Difficulty Level	Classifies notes as Beginner, Intermediate, or Advanced
Topic Information	Supports note searching and recommendations
Student Ratings	Records feedback on shared academic notes

Table 3.1 summarises the main data fields considered in the ANIE component for organising, evaluating, and recommending academic learning materials.

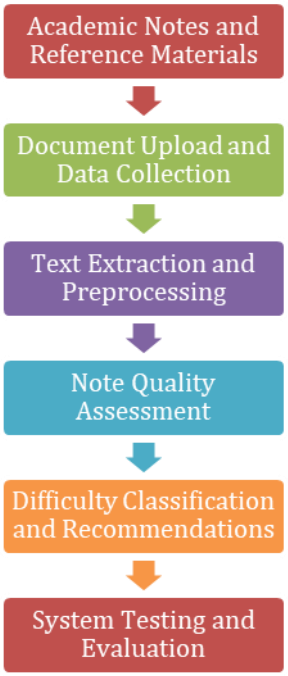


Figure 3.1: Academic Note Processing and Evaluation Workflow of ANIE

Figure 3.1 illustrates the main workflow of the Academic Note Intelligence Engine (ANIE). The process begins with academic note collection and document processing, followed by quality assessment, difficulty classification, and resource recommendations. Finally, the developed component is evaluated through system testing to examine its functionality and performance.

3.3 Experimental or study procedure

The study involved building ANIE and checking its operation within the ShareMate application. The workflow moved from uploading a note to processing its content, assessing its quality, identifying difficulty and suggesting related material.

Step 1 – Note Uploading: Academic notes in PDF and image formats were uploaded to the system for processing and evaluation.

Step 2 – Document Processing: The uploaded documents were processed to extract and organise academic content for further analysis.

Step 3 – Quality Assessment: ANIE evaluated notes based on relevance, completeness, clarity, and readability to generate quality scores.

Step 4 – Difficulty Classification: Notes were classified into Beginner, Intermediate, and Advanced levels based on their content.

Step 5 – Note Recommendations: Relevant academic notes were recommended based on course topics and learning requirements.

Step 6 – System Testing: The developed component was tested through the ShareMate application to examine its functionality, note assessment, recommendations, and document accessibility.

3.4 ANIE quality-scoring method

The Academic Note Intelligence Engine (ANIE) uses a weighted scoring approach to evaluate the quality of student-uploaded academic notes. The assessment considers five main criteria: conceptual correctness, clarity, reference validity, completeness, and readability. Each criterion contributes to an overall quality score that helps students understand the reliability and usefulness of shared learning materials.

The quality score is calculated using the following formula [S1]:

[
Q_{note} = \sum_{k=1}^{m} w_k S_k
]

Where:

- (Q_{note}) = Overall note-quality score
- (S_k) = Score assigned to each quality criterion
- (w_k) = Weight assigned to each criterion
- (m) = Total number of evaluation criteria

The resulting score provides a structured way to compare academic notes based on their quality. When reference materials are unavailable, the assessment should clearly indicate any limitations rather than treating the note as fully verified. This distinction makes the assessment more useful without presenting an automated score as lecturer approval.

3.5 Difficulty classification and recommendation

The Academic Note Intelligence Engine (ANIE) was developed to classify academic notes into three difficulty levels: Beginner, Intermediate, and Advanced. This classification considers the complexity of the content, technical terminology, and background knowledge required to understand the material. By identifying the difficulty level of each note, the system helps students choose learning resources that match their knowledge and learning needs.

ANIE also provides intelligent note recommendations by considering course relevance, related topics, note quality, and difficulty levels. This matters because a popular note, or one with matching keywords, is not necessarily the best choice for every learner. The intention is to make finding a suitable note less dependent on trial and error.

A conceptual recommendation ranking formula is expressed as:

$$R(n, u) = \alpha T + \beta Q + \gamma L + \delta F$$

where (R(n,u)) represents the recommendation score for note (n) and student (u), while (T), (Q), (L), and (F) represent topic relevance, note quality, difficulty suitability, and student feedback, respectively. The coefficients (\alpha), (\beta), (\gamma), and (\delta) represent the relative importance of each factor.

Together, difficulty classification and personalised recommendations improve the organisation and discovery of academic notes, helping students access learning materials more efficiently.

3.6 Measures, baselines and evaluation metrics

The performance of the Academic Note Intelligence Engine (ANIE) was assessed by considering note-quality evaluation, difficulty classification, recommendation relevance, and overall system functionality.

For note-quality assessment, the agreement between automated scores and human ratings can be measured using Spearman's rank correlation and Mean Absolute Error (MAE). Difficulty classification performance is measured using accuracy, precision, recall, F1-score, and a confusion matrix to evaluate how effectively notes are categorised into Beginner, Intermediate, and Advanced levels.

The recommendation mechanism is evaluated using Precision@K, Recall@K, and nDCG@K to measure the relevance and ranking of suggested notes. System performance is assessed through response time, successful note uploads, API reliability, and usability observations.

These evaluation measures provide a structured way to examine the effectiveness of ANIE in assessing academic notes and recommending suitable learning resources within the ShareMate platform.

Task	Baseline	Main metric	Additional checks
Quality assessment	Manual rubric / no ranking	Rank agreement	Per-criterion errors
Difficulty Classification	Majority class / TF-IDF model	Macro F1	Confusion matrix
Recommendation	Keyword-only /	nDCG@5	Precision@5, coverage

Task	Baseline	Main metric	Additional checks
	popularity		
User workflow	Manual note discovery	Task completion	usability survey
API performance	Controlled simple request	p95 latency	Error/timeout rates

Table 3.2: Evaluation Metrics and Baseline Comparison

Table 3.2 summarises the evaluation methods considered for the ANIE component. These measures provide a basis for assessing note quality, difficulty classification, recommendation relevance, usability, and system performance. Comparing the component with simpler baseline methods also helps identify its strengths and areas for improvement.

3.7 Ethics, privacy, security and responsible AI

Because ANIE handles material shared by students, privacy and academic integrity need attention throughout its design. Students' personal information must be protected, and the rights of note authors and lecturers must be respected. Lecturer-provided resources should be used with appropriate permission, and access to shared documents should be managed carefully.

Automated assessment also has practical limits. ANIE's quality score is guidance for choosing a note; it is not a guarantee that every explanation in the note is correct. It is also important to discourage plagiarism, misleading information, and unfair use of the reward system.

Safe handling of uploaded files is another requirement of a dependable sharing service. Uploaded documents need to be handled safely, and access to academic resources should be limited to authorised users. Taken together, these safeguards are intended to make the service more trustworthy for its student users.

3.8 Reproducibility and research artefacts

Reproducibility was an important consideration in the development and evaluation of the Academic Note Intelligence Engine (ANIE). Keeping clear records of the data, development process, and testing methods helps ensure that the research can be reviewed and repeated when necessary.

The research artefacts cover the note data, preprocessing steps, model settings where applicable, evaluation records and API documentation. Keeping software versions and test procedures together makes later review and repetition more practical.

Screenshots and testing evidence from the ShareMate application are included to demonstrate the implemented features and support the research findings. These materials provide a useful reference for understanding the development, testing, and evaluation of the ANIE component.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

ANIE was designed around a straightforward student need: uploading a note and finding other notes worth reading. Its core requirements covered PDF and image uploads, course and topic browsing, and access to relevant learning resources.

The design also includes note-quality assessment, difficulty labels and recommendations. Students can view quality scores, identify notes as Beginner, Intermediate, or Advanced, and discover materials that match their learning needs.

Additional requirements included student ratings, token-based rewards, and secure access to shared resources. Together, these features aim to encourage useful contributions without making the sharing process difficult to use.

The functional and non-functional requirements of ANIE are presented in the following tables.

ID	Functional requirement	Acceptance idea
FR01	Allow students to upload academic notes	PDF and image files can be uploaded and stored
FR02	Extract content from uploaded documents	Text is extracted or an error message is displayed
FR03	Compare notes with lecturer materials	Notes are matched with authorised reference content
FR04	Generate note-quality scores	Quality scores and assessment criteria are displayed
FR05	Classify notes by difficulty level	Notes are labelled Beginner, Intermediate, or Advanced
FR06	Search and recommend relevant notes	Suggestions are provided based on topics and difficulty
FR07	Allow students to rate and report notes	Ratings and reports are recorded for review
FR08	Provide rewards for useful contributions	Tokens are awarded according to contribution rules
FR09	Connect ANIE with the ShareMate application	API requests and responses are handled correctly
FR10	Allow students to manage uploaded notes	Students can edit or delete their own submissions

Dimension	Proposed non-functional requirement
Security	Authentication, scoped authorisation, safe file handling
Privacy	Minimal data retention, consent, deletion
Performance	Measure upload/process latency and p95 API time; final SLO to confirm

Dimension	Proposed non-functional requirement
Explainability	Show criterion-level assessment and missing references
Maintainability	Versioned endpoints; modular extraction/scoring/recommendation
Accessibility	Readable status, screen-reader labels and understandable messages
Scalability	Independent services and asynchronous processing for large PDFs
Reliability	Retry transient failures, durable ingestion state and audit logs

Table 4.1: Functional Requirements of the ANIE Component

Table 4.1 presents the functional requirements of the ANIE component, covering academic note management, automated assessment, difficulty classification, recommendations, student feedback, and integration with the ShareMate application.

4.2 Component architecture within ShareMate

The source paper describes an API-mediated service architecture joining FLTS, ANIE, FriendZone Ride and Grocery. It contains inconsistent high-level stack descriptions (its abstract mentions ReactJS/Node.js/MongoDB, while methodology describes Flutter/Python/Firebase). Accordingly, the final implementation stack must be confirmed from the actual ANIE repository and deployment environment rather than inferred. [S1]

A defensible component design separates mobile UI, authenticated API, file storage, ingestion/extraction worker, note-quality scorer, difficulty classifier, topic index, recommendation service, token/feedback service and persistent metadata store.

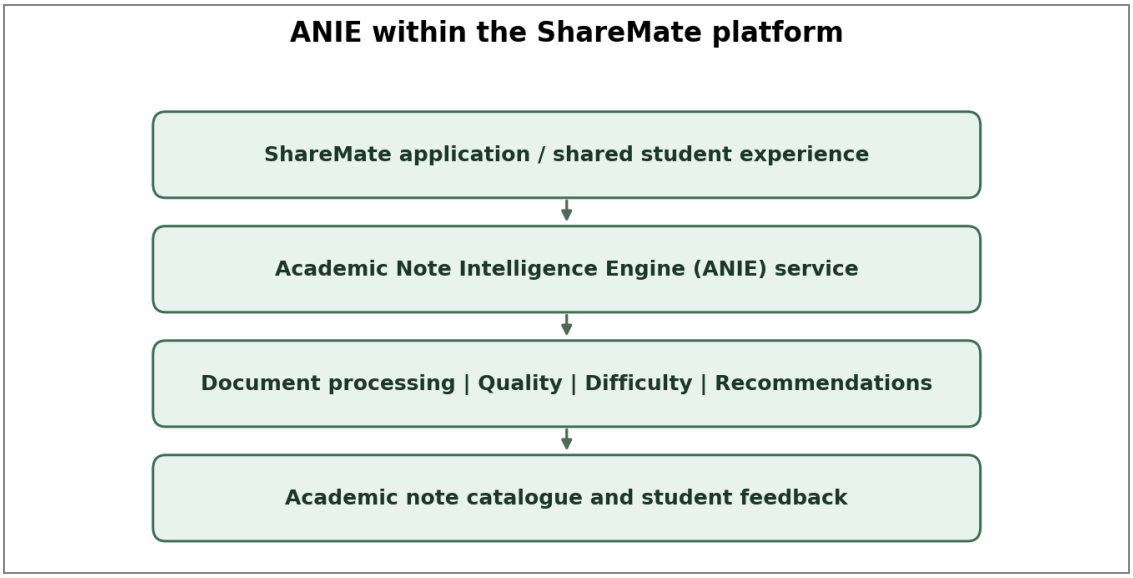


Figure 4.1. Overall ShareMate architecture showing four component services and ANIE in context



Figure 4.2. ANIE pipeline: upload → extract → compare → assess quality → classify → recommend → reward

Figure 4.2 illustrates the main processing pipeline of the Academic Note Intelligence Engine (ANIE). The process begins when a student uploads an academic note. The system then extracts the document text, compares the content with relevant lecturer-provided materials, and performs quality assessment. After that, the note is classified according to its difficulty level, and suitable recommendations are generated for students. Finally, feedback and reward-related actions support student participation and contribution.

4.3 Design alternatives and decisions

Reference comparison may use lexical matching, embedding similarity or rubric-aware evaluation; no single similarity score can prove factual correctness. Rule-based classifiers are interpretable but may miss nuanced meaning; supervised text models may be more flexible but need labelled data. Popularity-only recommendations risk amplifying low-quality notes; content-first ranking should include quality and level safeguards.

Design decisions should be made through documented experiments and practical considerations. The project-level paper specifies ANIE functions, but not the exact final classification model or hyperparameters.

4.4 Detailed design: users and sequence

Primary roles are Student Uploader, Student Reader, Lecturer/Moderator and Platform Administrator. A student selects course/topic, uploads a note, waits for validation and receives feedback. Another student searches for a topic, filters by desired difficulty, opens recommended content and optionally provides ratings or reports errors.

Each note should transition through states such as UPLOADED, PROCESSING, NEEDS_REVIEW, PUBLISHED and REJECTED/REMOVED. An uploader should not see an invented result while analysis is pending.

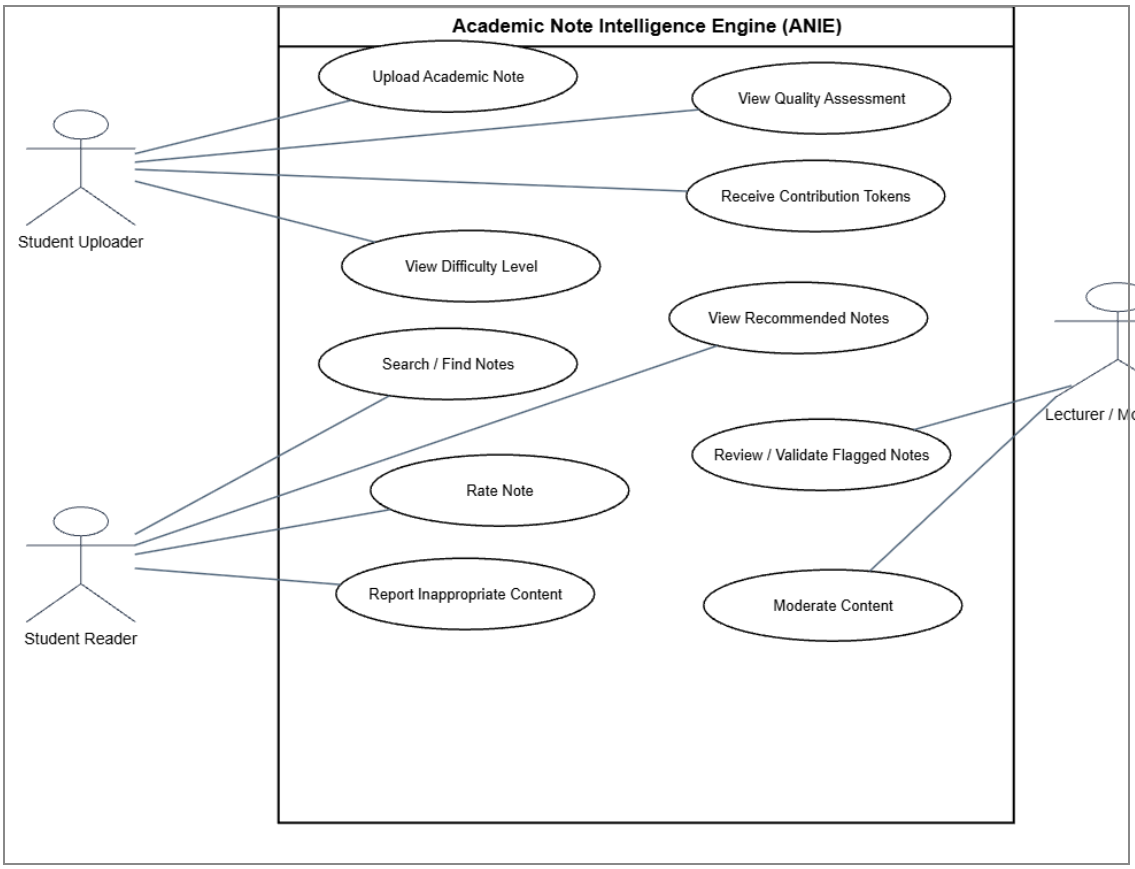


Figure 4.3. ANIE actors and use cases: upload, validate, find, rate, moderate and receive tokens

4.5 Data model and interfaces

Suggested entities are users, courses, modules, lecturer_references, notes, note_versions, extraction_results, quality_assessments, difficulty_predictions, topic_tags, recommendation_events, ratings, reports and token_transactions. Enforce ownership and course-based access permissions.

The following routes illustrate an API contract; substitute actual tested routes before final submission.

Method	Proposed route	Purpose
POST	/api/notes/upload	Submit file and course metadata
GET	/api/notes/{id}/status	Processing state
GET	/api/notes/{id}/assessment	Quality breakdown
GET	/api/notes/{id}/difficulty	Level and explanation
GET	/api/notes/search	Course/topic query
GET	/api/recommendations	Rank notes for topic/level
POST	/api/notes/{id}/feedback	Rating, report or correction
GET	/api/rewards/me	Contribution summary

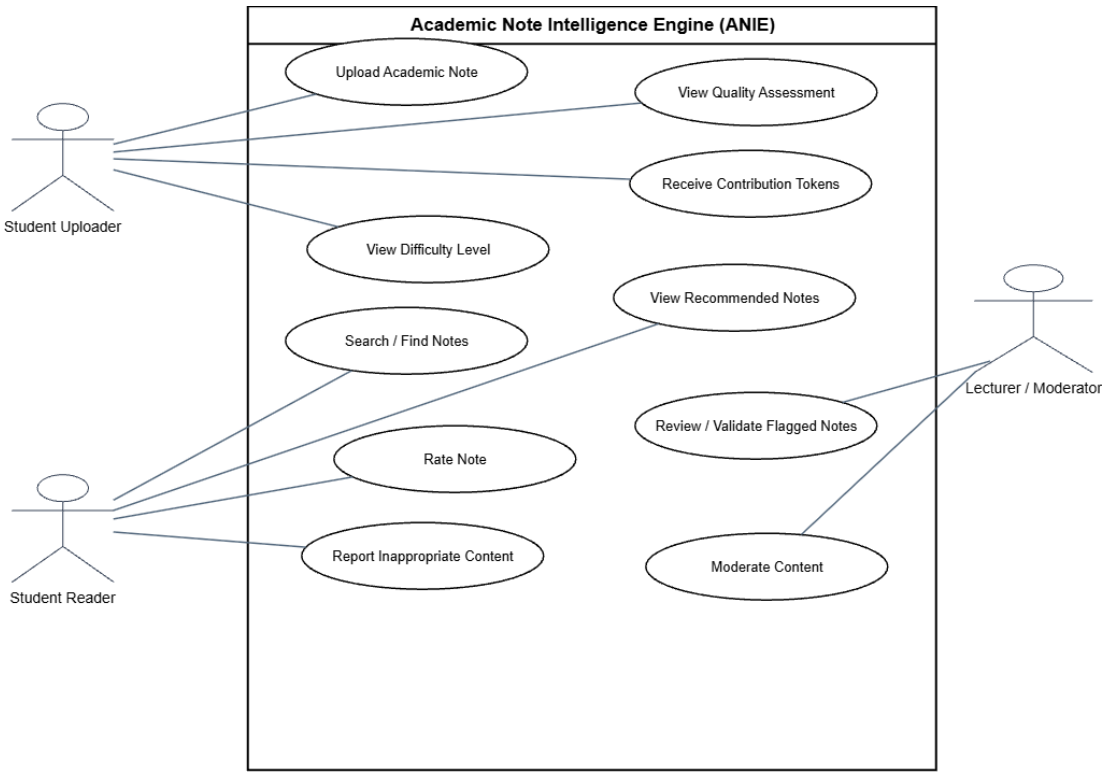


Figure 4.3 illustrates the main actors and their interactions with the Academic Note Intelligence Engine (ANIE). Student uploaders can share academic notes, view quality assessments and difficulty levels, and receive contribution tokens. Student readers can search for notes, access recommendations, provide ratings, and report inappropriate content. Lecturers or moderators can review flagged notes and moderate shared academic resources.

4.6 Standards and professional practice

Follow institutional research ethics, intellectual-property rules, accessibility practices, secure API development principles and reproducible ML reporting. Adopt clear failure messages, version-control policies and moderation escalation for potentially incorrect educational information.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The ShareMate project document states a microservices-style architecture with mobile interface and cloud-backed services [S1]. Because the uploaded paper gives more than one stack description, the implementation chapter must identify technologies directly from the current ANIE source code. The entries below are verification prompts, not asserted installations.

Layer	Candidate/source context	Final evidence needed
Frontend	Flutter stated in methodology; ReactJS stated in abstract	Package manifest / screenshots
Backend	Python stated in methodology; Node.js/Express stated in abstract	Actual ANIE service repository
Data services	Firebase stated in methodology; MongoDB in abstract	Actual schema/collections
Parsing/OCR	To be documented	Dependencies and test data
Models	To be documented	Artifact names, versions, experiments
API exposure	Ngrok for development noted in paper	Real endpoint demonstration

5.2 Component implementation workflow

Upload handler: accept a file only from an authorised user, validate MIME type and extension, generate a safe storage key, create a metadata record and enqueue analysis. Return a processing identifier rather than holding open a long request for large scans.

Content-processing worker: extract text, identify languages and headings, tag topic candidates, retain extraction-quality indicators and block downstream scoring when the document is unreadable. Scoring service: retrieve permitted lecturer reference content, compute criterion evidence and clearly distinguish automatic similarity from verified conceptual correctness.

Level classification service: classify and attach reason indicators; publish confidence or fall back to manual review when evidence is insufficient. Recommender: filter by course and available permission, score topic relevance, quality and difficulty

preference; record which version of the ranking logic generated the recommendations.

Reward service: award tokens only after passing quality/duplication and moderation checks. Include reversible reward transactions so a later plagiarism or abuse finding can revoke inappropriate credit.

5.3 Proposed processing algorithm

Algorithm ANIE-Process(note, course): (1) authenticate uploader; (2) validate file and permission; (3) extract text; (4) if unreadable, return NEEDS_REVIEW; (5) identify course topics; (6) load authorised reference units; (7) compute criterion evidence and normalised score; (8) classify difficulty; (9) save assessment and explanation; (10) update recommendation index; (11) determine publication/moderation state; (12) evaluate eligibility for token award.

Algorithm ANIE-Recommend(user, topic): (1) restrict to permitted notes; (2) retrieve relevant candidates; (3) exclude flagged or low-confidence items as policy requires; (4) calculate topic, quality and difficulty fit; (5) rank and diversify; (6) expose understandable reason labels; (7) log evaluation feedback without recording unnecessary personal data.

5.4 Interface implementation and screenshot evidence

The following screenshots present the implemented interfaces of the Academic Note Intelligence Engine (ANIE) within the ShareMate application.

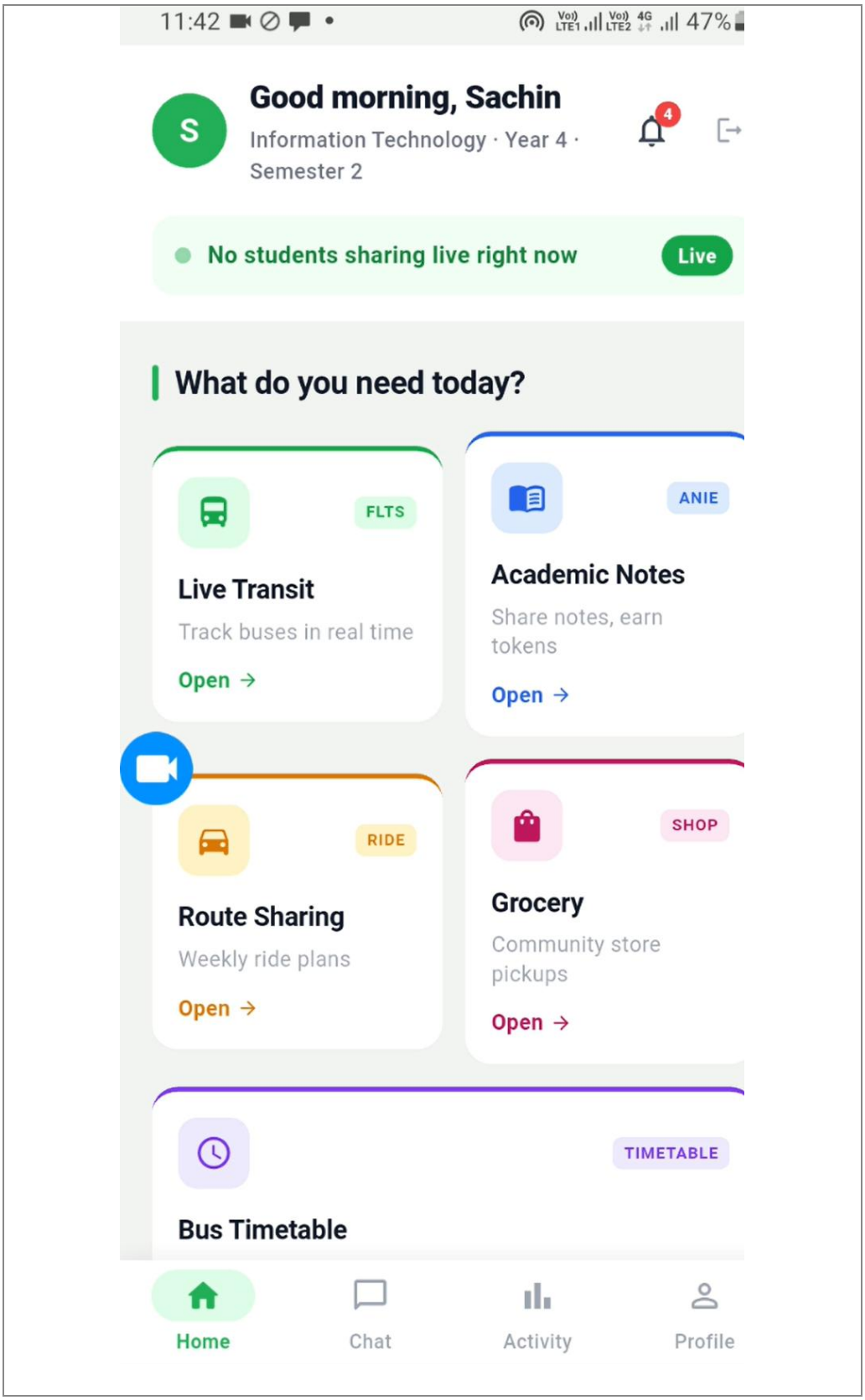


Figure 5.1. ShareMate home screen

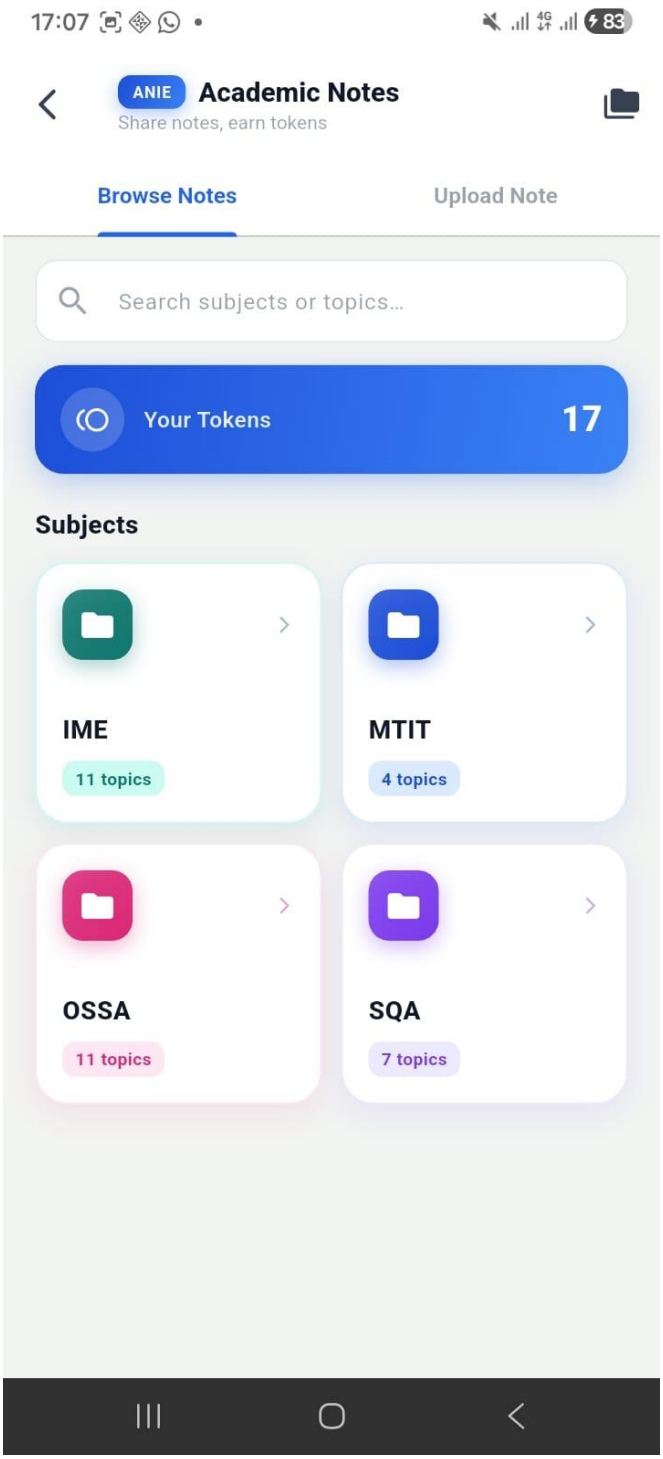


Figure 5.2. ANIE notes browse screen

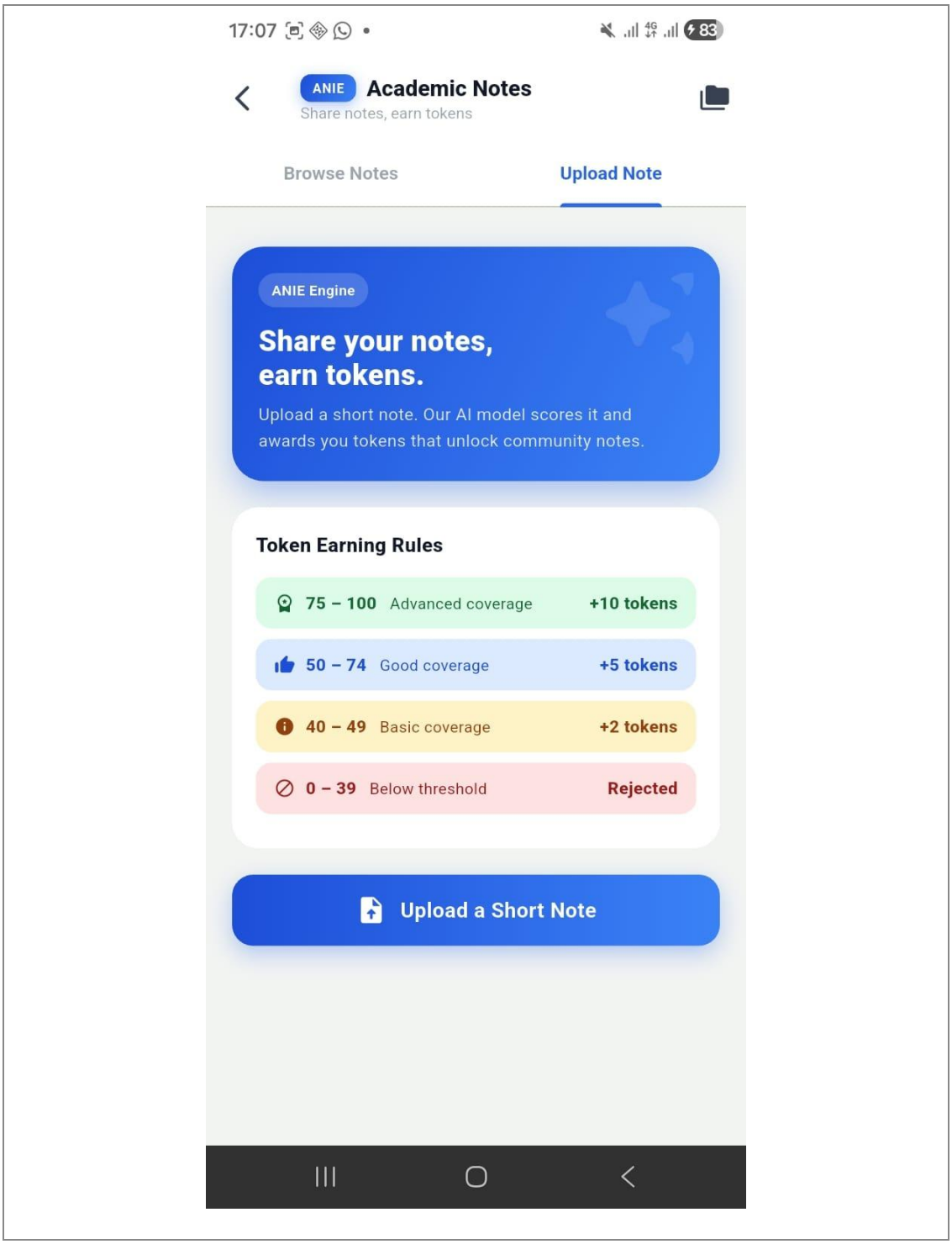


Figure 5.3. ANIE upload note overview screen

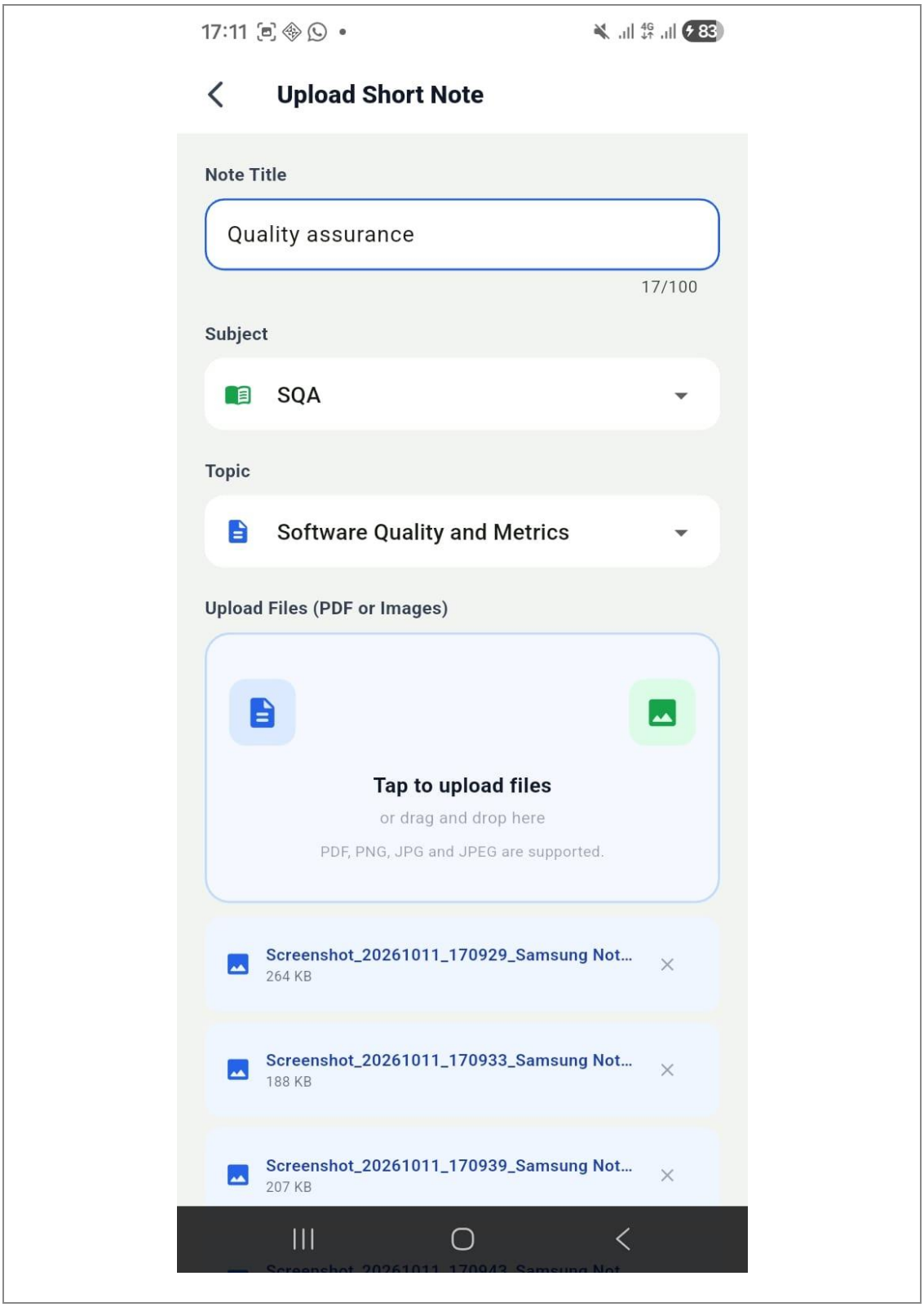


Figure 5.4. Upload academic note form

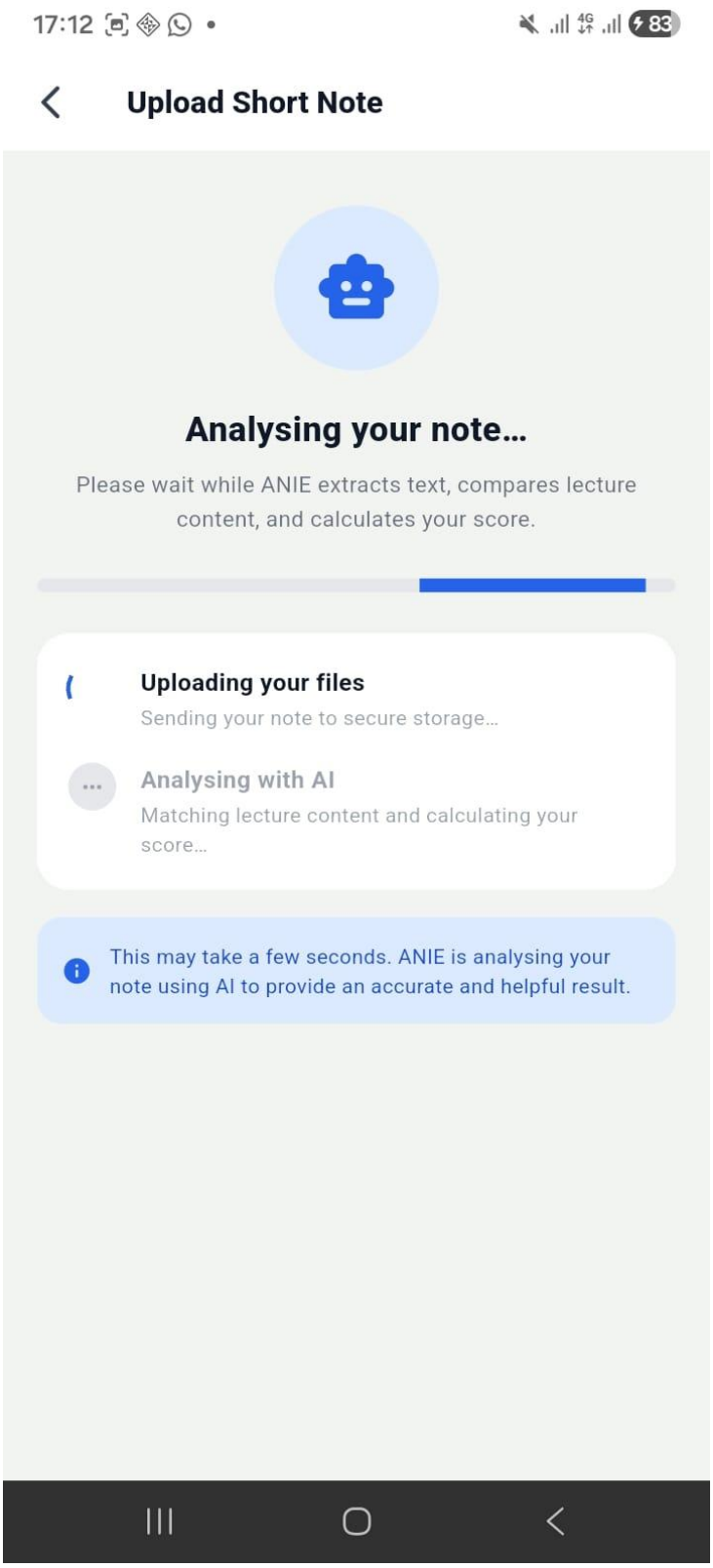


Figure 5.5. Academic note analysis processing screen

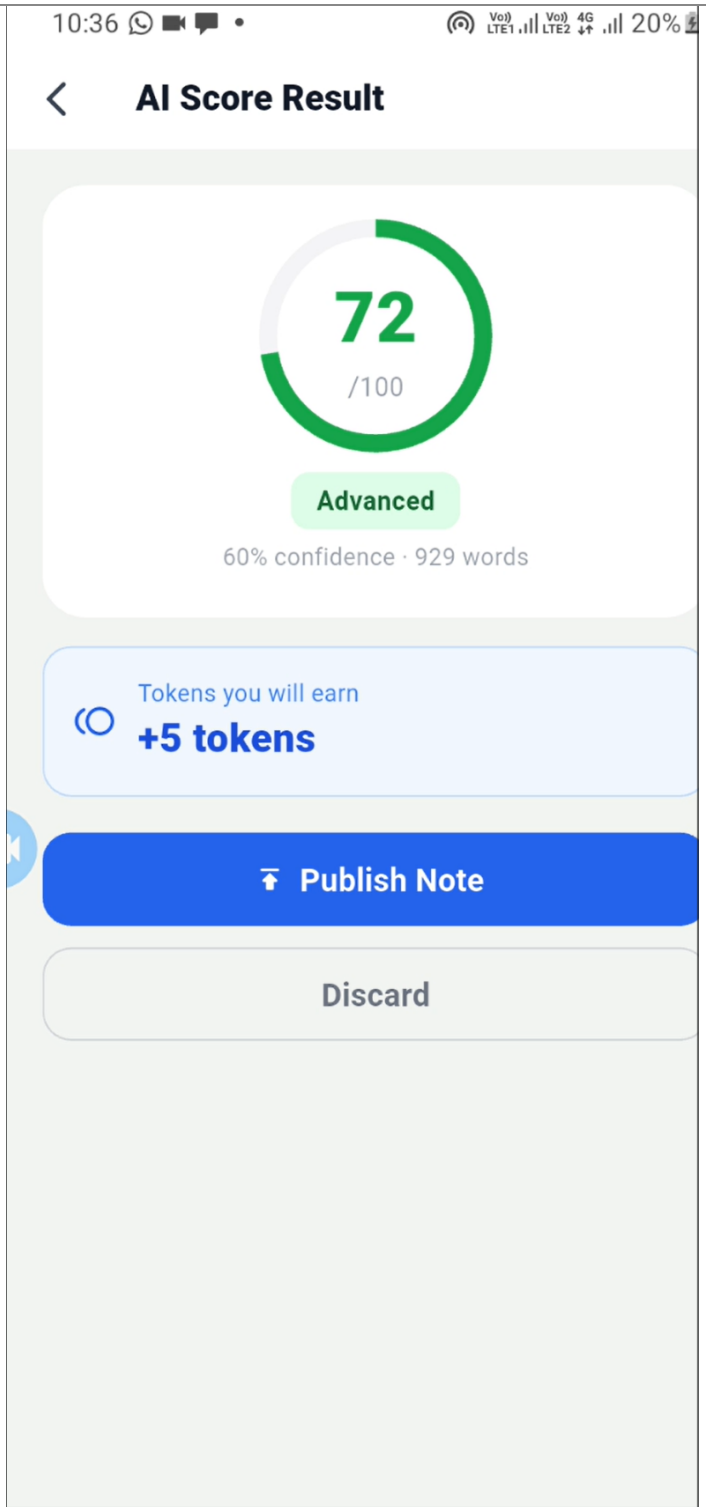


Figure 5.6. Quality assessment and difficulty classification result

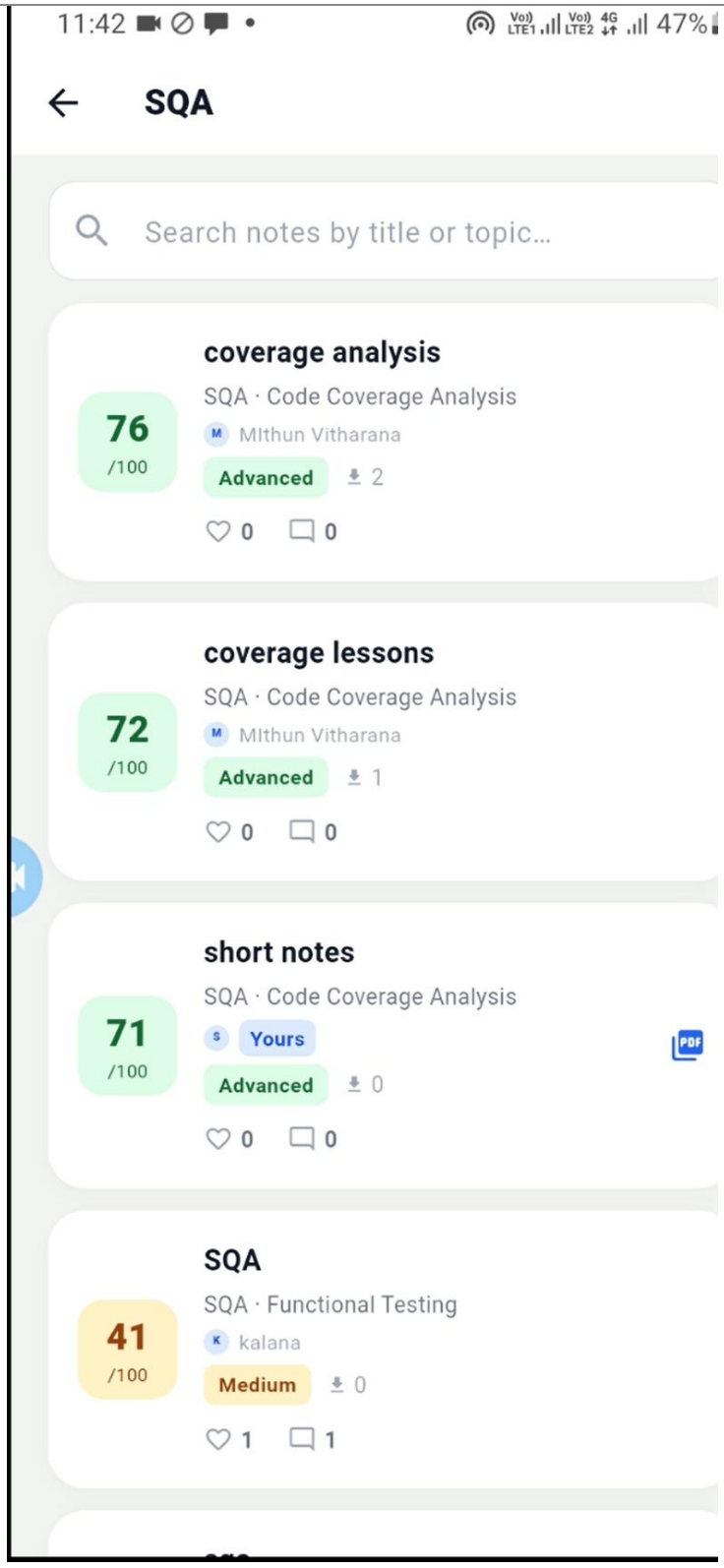


Figure 5.7. Subject-based recommended note listing

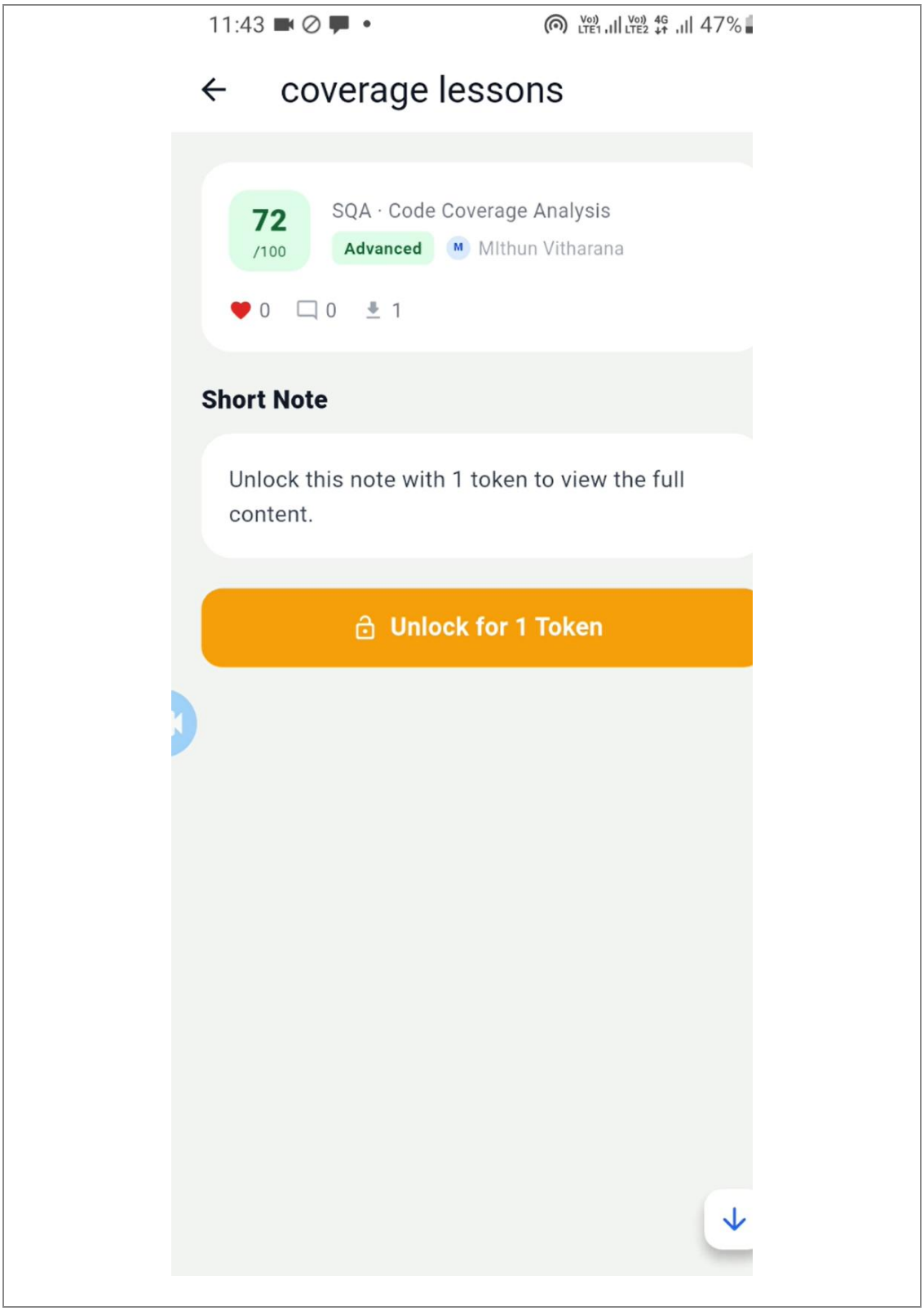


Figure 5.8. Token-based note access screen

5.5 Integration with other ShareMate components

ANIE should receive a verified user identity and course context from the common ShareMate application and return only appropriately scoped note metadata, assessment outcomes, content links and recommendation cards. Cross-service interaction must use authenticated APIs. Update token balances or common profile views through a consistent, auditable mechanism.

Integration tests should demonstrate successful login-to-upload navigation, correct user permissions, safe deep links to notes, and accurate propagation of assessment and token events. A service outage in ANIE should not break unrelated transportation or grocery components.

5.6 Verification and software testing

Use a combination of unit tests for scoring and ranking logic, file-parser tests, API contract tests, security tests, integration tests, load tests and acceptance testing with students. Test with intentionally misleading, duplicate, image-only, corrupted and mixed-topic notes. Capture evidence of both success and graceful failure.

Table 5.1:Functional Testing Results of the ANIE Component

Test ID	**Test Scenario**	**Expected Result**	**Actual Result**
TC01	Academic Note Uploading	Academic notes can be uploaded successfully	Academic note uploaded and processed successfully
TC02	Document Processing	Uploaded document content is processed	Document processing completed and assessment generated
TC03	Note Quality Assessment	System generates a quality score	Quality score generated and displayed successfully

TC04	Subject-Based Note Browsing	Notes are organised according to subjects	Academic notes displayed under the selected subjects
TC05	Academic Note Viewing	Students can open and read selected notes	Selected academic document opened successfully
TC06	Token-Based Note Access	Students can access notes using tokens	Token-based note unlocking functioned successfully

Table 5.1 presents the functional testing results of the Academic Note Intelligence Engine (ANIE) within the ShareMate application. The results show the operation of the main features, including academic note uploading, document processing, quality assessment, subject-based browsing, document viewing, and token-based access. These functions support a more organised and accessible academic note-sharing experience for university students.

5.7 Deployment and operation

The Academic Note Intelligence Engine (ANIE) was integrated into the ShareMate application to support academic note uploading, automated quality assessment, difficulty classification, and resource recommendations. The system follows a service-based architecture, allowing the academic note-sharing component to communicate with the main application through APIs.

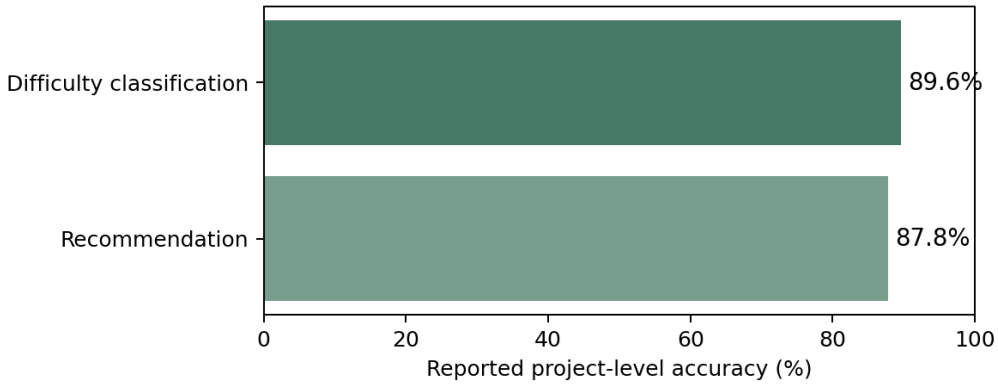
The ShareMate project uses a mobile application supported by backend services and cloud-based data management. During development and testing, the integration process focused on ensuring that academic notes could be uploaded, processed, stored, and accessed through the application. The component also supported note browsing, quality score display, and token-based access to learning materials.

Operational considerations included maintaining reliable communication between services, managing uploaded academic resources, and protecting user information. These measures are important for supporting the continued operation and maintenance of the ANIE component within ShareMate.

6 RESULTS AND VALIDATION

6.1 Evidence boundaries

The ShareMate research paper reports two main performance results for the Academic Note Intelligence Engine (ANIE): 89.6% accuracy in difficulty classification and 87.8% accuracy in academic note recommendations [S1]. These results provide an indication of the component's performance in supporting intelligent academic resource sharing.



Source: ShareMate project paper [S1]. Protocol not independently documented.

Figure 6.1: Reported ANIE Classification and Recommendation Performance

Figure 6.1 presents the reported classification and recommendation results. Although these findings are encouraging, the research paper does not provide detailed information about the dataset, testing procedure, or evaluation metrics used to obtain them. Therefore, the values are presented as reported project-level results rather than independently reproduced experimental findings.

Table 6.1: Reported Performance Results of ShareMate

Evaluation Metric	**Reported Result**	**Interpretation**
ANIE Difficulty Classification Accuracy	89.6%	Reported in the ShareMate research paper
ANIE Recommendation Accuracy	87.8%	Reported in the ShareMate research paper

Overall System Response Time	420 ms	Average response time of the complete ShareMate platform

Table 6.1 presents the performance results reported in the ShareMate research paper. The Academic Note Intelligence Engine (ANIE) recorded a classification accuracy of 89.6% and recommendation accuracy of 87.8%. The overall ShareMate platform reported an average response time of 420 milliseconds [S1]. These findings provide an overview of the reported system performance in supporting academic resource sharing.

6.2 Individual research results

The Academic Note Intelligence Engine (ANIE) was developed and tested as part of the ShareMate application. The evaluation focused on the main functions of the component, including academic note uploading, document processing, automated quality assessment, and access to shared learning materials.

During functional testing, the application supported academic note uploads, displayed generated quality scores, and allowed students to browse notes according to their subjects. Students could also open selected documents and access learning materials through the token-based system.

These results show that the main note-sharing functions were operational within the ShareMate application. The classification and recommendation performance reported in the overall ShareMate research paper is presented separately in Section 6.1.

Table 6.2: Individual ANIE Component Evaluation Summary

Evaluation Area	ANIE Result
Academic Note Uploading	PDF note uploading and processing completed
Document Processing	Uploaded academic content processed
Note Quality Assessment	Automated quality score generated and displayed
Subject-Based Note Browsing	Notes displayed according to selected subjects
Academic Note Viewing	Selected academic documents opened successfully
Token-Based Access	Token-based note unlocking supported

Table 6.2 summarises the main functional outcomes of the developed Academic Note Intelligence Engine (ANIE). The component supported academic note uploading, document processing, automated quality score generation, subject-based browsing, document viewing, and token-based note access. These results highlight the practical operation of ANIE within the ShareMate application and its contribution to improving academic resource sharing.

6.3 Qualitative validation and error analysis

Qualitative validation focused on reviewing the main functions of the Academic Note Intelligence Engine (ANIE) and identifying factors that could affect the reliability of its results. The component was examined in relation to academic note processing, quality assessment, difficulty classification, and resource recommendations.

The quality assessment feature generates a score for uploaded academic notes. However, the reliability of this score may be affected by incomplete content, unclear explanations, or differences between student notes and lecturer-provided materials. These factors highlight the importance of accurate document processing and appropriate reference materials.

For difficulty classification, distinguishing between Beginner, Intermediate, and Advanced notes can be challenging when a document contains content from more than one difficulty level. Similarly, note recommendations may be less relevant when different academic topics use similar terminology or when limited learning materials are available for a particular subject.

The implemented application also provides students with access to academic notes through subject-based browsing and document viewing. These features support the practical use of the system, although a separate user study would be required to evaluate student satisfaction and the overall usability of the recommendations.

Overall, the analysis identifies potential limitations of automated academic note evaluation and recommendation. Further testing with lecturer-reviewed materials and

student feedback would help establish the reliability and usefulness of ANIE in different academic contexts.

6.4 Product and integration outcomes

The Academic Note Intelligence Engine (ANIE) was integrated into the ShareMate application to support academic note sharing and evaluation. The integration allows students to access academic resources through the application's interface and interact with the note assessment features.

The implemented workflow supports academic note uploading and processing, followed by the display of a generated quality score. Students can also browse available notes according to their subjects, open selected documents, and access learning materials through the token-based system.

These features demonstrate how ANIE contributes to the overall ShareMate platform by combining academic note assessment with resource accessibility. The integration provides a practical connection between automated note evaluation and the student-facing note-sharing experience.

6.5 Summary of findings

The evaluation of the Academic Note Intelligence Engine (ANIE) highlights its contribution to intelligent academic resource sharing within the ShareMate application. The ShareMate research paper reports a difficulty classification accuracy of 89.6% and a recommendation accuracy of 87.8% [S1].

At the application level, ANIE supports academic note uploading, document processing, automated quality score display, subject-based browsing, document viewing, and token-based access. These functions provide students with a structured way to share and access academic learning materials.

However, the reported performance figures do not include detailed individual experimental results or baseline comparisons. Further evaluation using independently documented test datasets and student feedback would provide a more complete assessment of the component's accuracy, reliability, and usability.

Overall, the findings indicate that ANIE provides a functional foundation for automated academic note assessment and resource discovery within ShareMate.

7 DISCUSSION

7.1 Interpretation of findings

The findings indicate that the Academic Note Intelligence Engine (ANIE) contributes to improving how students evaluate and access shared academic notes within the ShareMate application. Unlike traditional note-sharing methods, ANIE combines automated quality assessment, difficulty classification, and academic resource recommendations to help students identify learning materials that are relevant to their needs.

The quality assessment feature provides students with an indication of the reliability and usefulness of uploaded notes. Difficulty classification into Beginner, Intermediate, and Advanced levels also aims to help students select materials that match their understanding of a subject. Together, these features support a more structured approach to academic note sharing.

The ShareMate research paper reports a difficulty classification accuracy of 89.6% and a recommendation accuracy of 87.8% [S1]. These results suggest the potential effectiveness of ANIE in supporting academic resource discovery. However, the available evaluation does not provide sufficient details about the dataset, testing procedures, or individual experimental comparisons to establish consistent performance across different academic subjects.

Overall, ANIE demonstrates the potential of combining academic note evaluation and personalised recommendations within a single platform. Further evaluation using diverse academic materials and student feedback would help determine its reliability and practical usefulness in different learning environments.

7.2 Comparison with related approaches

Traditional academic note-sharing platforms mainly focus on uploading, storing, and accessing learning materials. Although these systems make resource sharing easier, they often provide limited support for evaluating note quality or identifying materials suitable for different learning levels.

The Academic Note Intelligence Engine (ANIE) addresses these limitations by combining automated note quality assessment, difficulty classification, and academic resource recommendations within the ShareMate application. Instead of relying only on file names, keywords, or popularity, ANIE aims to support more informed resource selection through quality scores and learning-level information.

Compared with recommendation systems that focus mainly on content similarity or user preferences, ANIE incorporates lecturer-provided reference materials into its proposed assessment process. This approach is intended to improve the relevance and reliability of shared academic notes. The token-based reward mechanism also encourages students to contribute learning materials to the platform.

However, the available evaluation does not establish whether each ANIE feature performs better than the corresponding traditional approach. A controlled comparison with keyword-based retrieval, popularity-based recommendations, and recommendation methods without quality assessment would provide stronger evidence of ANIE's effectiveness.

Overall, ANIE offers an integrated approach to academic note sharing by bringing quality evaluation, difficulty classification, resource discovery, and student participation together within a single platform.

7.3 Individual research contribution

The main individual research contribution of this study is the development and integration of the Academic Note Intelligence Engine (ANIE) within the ShareMate application. The component brings together academic note quality assessment, difficulty classification, and resource recommendation to support more effective academic note sharing among university students.

A key contribution is the use of lecturer-provided reference materials in the proposed note evaluation process. This approach aims to assess academic notes based on their relevance, accuracy, and overall usefulness rather than relying only on file names or popularity. The classification of notes into Beginner, Intermediate, and Advanced levels also helps students identify materials that are suitable for their learning needs.

Another contribution is the integration of these features into a student-centred application. ANIE connects note uploading and assessment with subject-based browsing, document access, and token-based participation, providing a structured workflow for sharing and discovering academic resources.

The contribution of this research lies primarily in combining these functions within a single academic note-sharing environment. Although the reported classification and recommendation results indicate promising performance, further independent evaluation is needed to establish how effectively the combined approach performs compared with existing methods.

7.4 Product contribution

ANIE contributes to the ShareMate platform by providing intelligent academic note assessment and resource recommendations. Its integration allows students to upload, evaluate, and access learning materials through a shared application. The modular design also supports future improvements without affecting other ShareMate components.

7.5 Limitations and threats to validity

ANIE has several limitations that may affect its accuracy and reliability. Limited lecturer reference materials, errors in text extraction, and differences in note formats can influence quality assessment. Difficulty classification may also be subjective, while recommendations may become less relevant when academic resources are limited.

Furthermore, automated quality scores cannot guarantee that every note is factually correct. Additional testing with diverse datasets and human reviewers is needed to evaluate the system's reliability.

7.6 Implications

This research highlights the potential of AI-assisted academic note sharing in higher education. Students can benefit from quality indicators and difficulty-based recommendations, while institutions may use similar approaches to organise learning

resources. Future improvements should focus on transparent evaluation, human feedback, and reliable academic content assessment.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The Academic Note Intelligence Engine (ANIE) is designed to support university students in finding and sharing suitable academic learning materials. Students often depend on notes shared by their classmates, but identifying reliable and understandable resources can be challenging. ANIE addresses this need by combining automated quality assessment, difficulty classification, and academic note recommendations within the ShareMate application.

The primary target users are university students who need access to relevant study materials. Students who contribute notes can also benefit from quality feedback and participation rewards. Lecturers and educational institutions represent potential stakeholders, particularly in providing reference materials and supporting responsible academic content sharing.

The main value of ANIE lies in helping students make better-informed decisions when selecting academic notes. Rather than browsing large collections of documents without guidance, students can use quality indicators and learning-level information to identify resources that may suit their academic needs.

8.2 Market and competitive position

Academic note-sharing services operate alongside learning management systems, cloud storage platforms, and online educational resource repositories. Although these platforms allow students to store and exchange learning materials, they may not provide automated support for evaluating note quality or identifying the difficulty level of individual documents.

ANIE aims to address this gap by bringing together note assessment, difficulty classification, and recommendation features within a single platform. Its integration with ShareMate also creates opportunities for students to access academic resources through an application that supports several other student-oriented services.

The combination of these features offers a potential advantage over conventional document-sharing approaches. However, the commercial demand for ANIE,

including market size, user willingness to pay, and institutional adoption, has not yet been formally evaluated. Further market research would be necessary before introducing the component as a commercial educational service.

8.3 Sustainability model

The long-term sustainability of ANIE depends on maintaining reliable academic content, supporting student participation, and managing the operational costs of the system. One possible approach is to collaborate with universities or educational institutions that may be interested in supporting digital academic resource sharing.

Institution-supported hosting could help cover expenses related to document storage, processing, and system maintenance. Partnerships with campus technology services may also provide opportunities to improve the platform and expand its availability to more students.

The token-based participation mechanism can encourage students to share useful learning materials and remain engaged with the platform. These tokens should primarily function as non-monetary participation rewards unless additional legal and financial considerations are addressed.

Sustainability also requires responsible content management. The platform should respect copyright ownership, discourage plagiarism, and ensure that academic materials are shared with appropriate permission. These practices would help establish trust among students, lecturers, and participating institutions.

8.4 Scalability, intellectual property and compliance

As the number of users and uploaded documents increases, ANIE may require improvements to its processing capacity, storage management, and overall system performance. Future development could introduce asynchronous document processing, improved caching, and more efficient management of computationally demanding tasks such as text extraction and quality assessment.

The system should also support secure access to academic resources and appropriate protection of student information. Uploaded notes may contain copyrighted material,

lecturer-provided content, or personal information, making responsible data handling an important requirement.

Students and institutions should retain their applicable ownership and usage rights over shared academic materials. Clear policies for content reporting, removal, user permissions, and data retention would be necessary for wider institutional deployment.

Addressing these technical and ethical considerations would support the future expansion of ANIE while maintaining reliability and responsible academic resource sharing.

8.5 Contribution to Sustainable Development Goals

ANIE primarily aligns with **Sustainable Development Goal 4 (Quality Education)** by supporting access to academic learning materials and encouraging knowledge sharing among university students. Through automated quality assessment, difficulty classification, and note recommendations, the system aims to help students identify resources that are relevant to their individual learning needs.

The component may also contribute to **Sustainable Development Goal 10 (Reduced Inequalities)** by creating opportunities for students with different levels of academic understanding to access shared learning resources. The Beginner, Intermediate, and Advanced classification approach is intended to make resource selection more accessible to students with varying knowledge levels.

By encouraging peer learning and responsible academic resource sharing, ANIE has the potential to support a more collaborative educational environment. However, these contributions represent the intended social and educational benefits of the system rather than independently measured Sustainable Development Goal outcomes.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research focused on the development of the Academic Note Intelligence Engine (ANIE) as part of the ShareMate application. The main aim was to improve how university students share, evaluate, and access academic learning materials. Traditional note-sharing methods often make it difficult for students to identify reliable notes or select resources that match their level of understanding.

ANIE addresses these challenges by bringing together academic note quality assessment, difficulty classification, and resource recommendations within a single platform. The application also supports note uploading, subject-based browsing, document viewing, and token-based access.

The ShareMate research paper reports a difficulty classification accuracy of 89.6% and a recommendation accuracy of 87.8% [S1]. These findings suggest the potential of ANIE to support academic resource discovery, although further independent testing is needed to establish its performance across different subjects and note formats.

Overall, this research provides a foundation for a more organised and intelligent academic note-sharing environment that encourages collaborative learning among university students.

9.2 Achievement of objectives

The research addressed its main objectives through the design and integration of ANIE within ShareMate. The implemented application supports academic note uploading and processing, automated quality score display, subject-based browsing, document viewing, and token-based access to learning materials.

The research also incorporated difficulty classification and academic note recommendation approaches intended to help students identify suitable learning resources. The reported project-level results provide an initial indication of their performance, while further individual evaluation would strengthen the evidence for these features.

Although the main system functions were developed, some objectives, particularly detailed model evaluation, human validation, and reproducibility, require additional experimental evidence to establish their effectiveness fully.

Table 9.1: Achievement of Research Objectives

Objective	**Achievement Status**	**Outcome**
O1 – Secure Note Intake	Partially Verified	Academic note uploading and processing were demonstrated; security testing requires further verification.
O2 – Quality Assessment	Functionally Verified	Automated quality scores were generated and displayed.
O3 – Difficulty Classification	Reported at Project Level	Classification accuracy of 89.6% was reported in the ShareMate research paper.
O4 – Academic Note Recommendations	Reported at Project Level	Recommendation accuracy of 87.8% was reported in the ShareMate research paper.
O5 – Token-Based Incentives	Partially Verified	Token-based note access was demonstrated; reward allocation requires further verification.
O6 – Application Integration	Functionally Verified	Academic note-sharing functions were accessible within ShareMate.
O7 – Reproducible Evaluation	Further Validation Required	Additional individual evaluation data and experimental documentation are required.

Table 9.1 summarises the achievement of the research objectives. The main user-facing functions of ANIE were implemented, while classification and recommendation performance were reported in the ShareMate research paper [S1]. Further independent evaluation is needed to assess the reliability and reproducibility of the component.

9.3 Individual research and product contributions

The main contribution of this research is the development and integration of the Academic Note Intelligence Engine (ANIE) within the ShareMate application. The component combines automated academic note quality assessment, difficulty classification, and resource recommendations to improve how students discover and access learning materials.

ANIE also supports a more organised academic note-sharing experience by connecting note evaluation with document uploading, subject-based browsing, and token-based access. Its integration into ShareMate demonstrates how intelligent academic resource management can be incorporated into a broader student-centred application.

The individual research contribution focuses on the ANIE component, while the overall ShareMate platform was developed as a collaborative team project.

9.4 Future directions

Future development of ANIE could focus on improving the accuracy, accessibility, and reliability of academic note assessment. Supporting Sinhala, Tamil, and English documents would allow more students to benefit from the system. Improved recognition of handwritten notes, mathematical equations, and diagrams could also strengthen the evaluation of different academic materials.

Other possible improvements include plagiarism detection, lecturer-assisted note verification, and more personalised recommendations based on students' learning needs. Offline document access and stronger privacy controls could further improve the user experience.

Future research should also evaluate ANIE using larger datasets, different academic subjects, and feedback from students and lecturers. These improvements would help determine its long-term effectiveness and support the development of a more reliable academic note-sharing platform.

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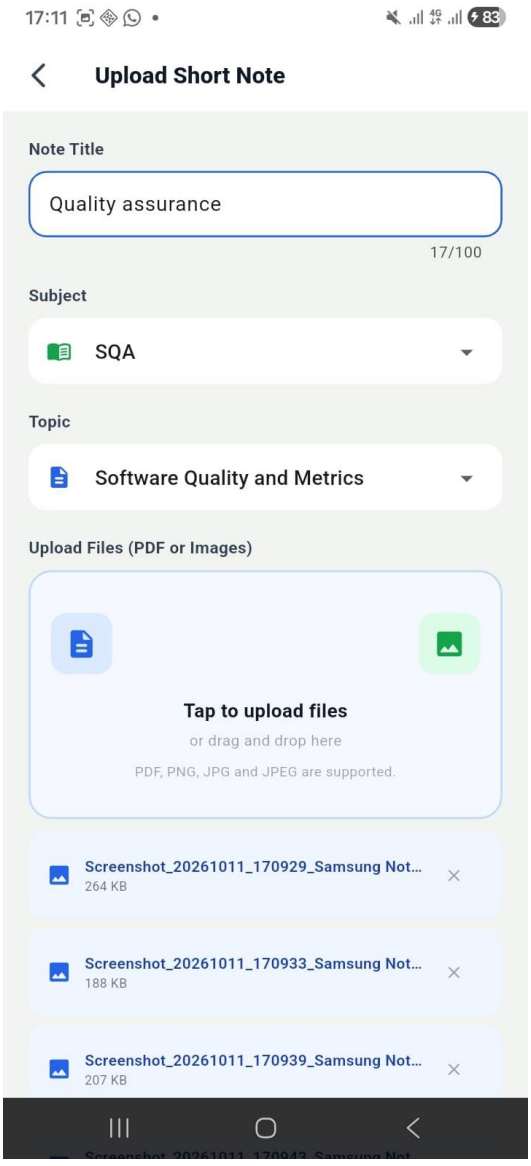
GLOSSARY

Term	Meaning
ANIE	Academic Note Intelligence Engine
OCR	Optical Character Recognition
NLP	Natural Language Processing
LMS	Learning Management System
API	Application Programming Interface
Quality Score	A weighted score used to evaluate academic note quality based on defined assessment criteria
Difficulty Class	The learning level assigned to a note: Beginner, Intermediate, or Advanced
Precision@K	The proportion of relevant recommendations among the top K suggested items
nDCG@K	A metric that evaluates the relevance and ranking of the top K recommended items
Macro F1-Score	The average F1-score calculated equally across all classification categories
Human-in-the-Loop	The involvement of human reviewers in checking or improving automated assessments

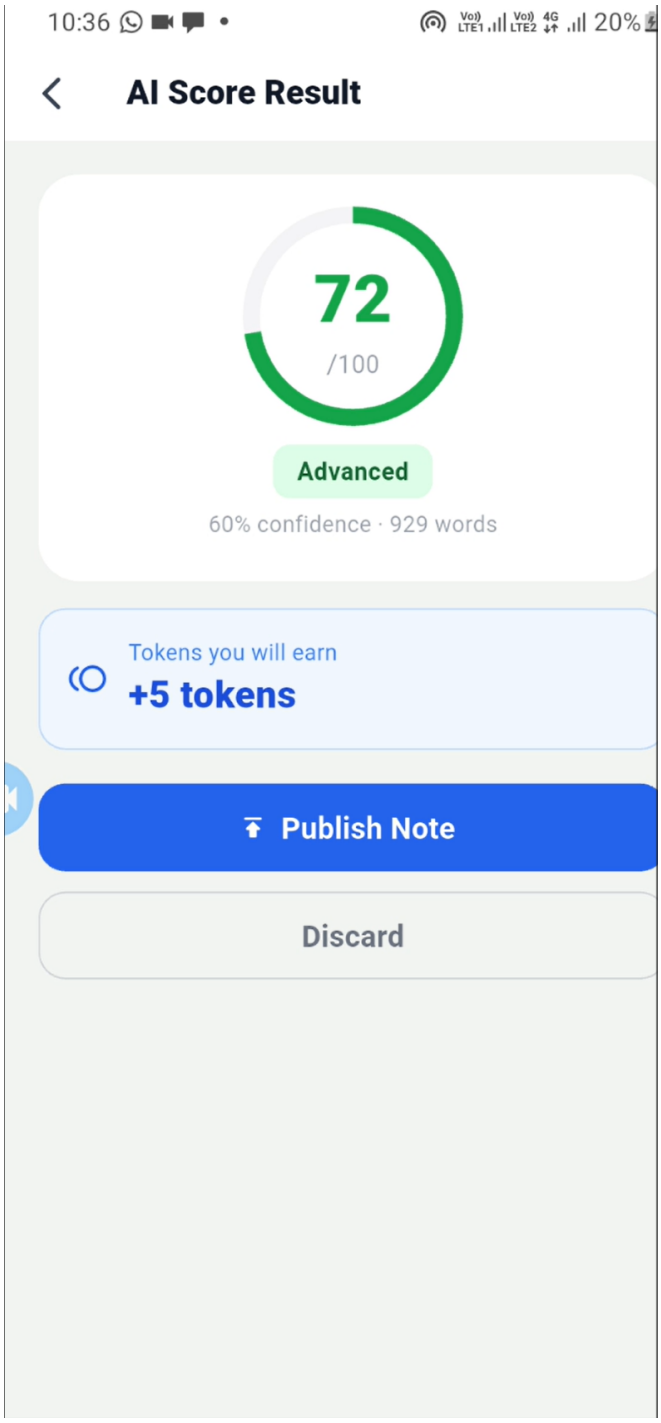
APPENDICES

Appendix A – ANIE System Implementation Screenshots

- This appendix presents selected implementation screenshots of the Academic Note Intelligence Engine (ANIE) within the ShareMate application.
- Figure A.1: Academic Note Upload Interface



- Figure A.2: Academic Note Quality Assessment Result



- Figure A.3: Subject-Based Academic Note Browsing

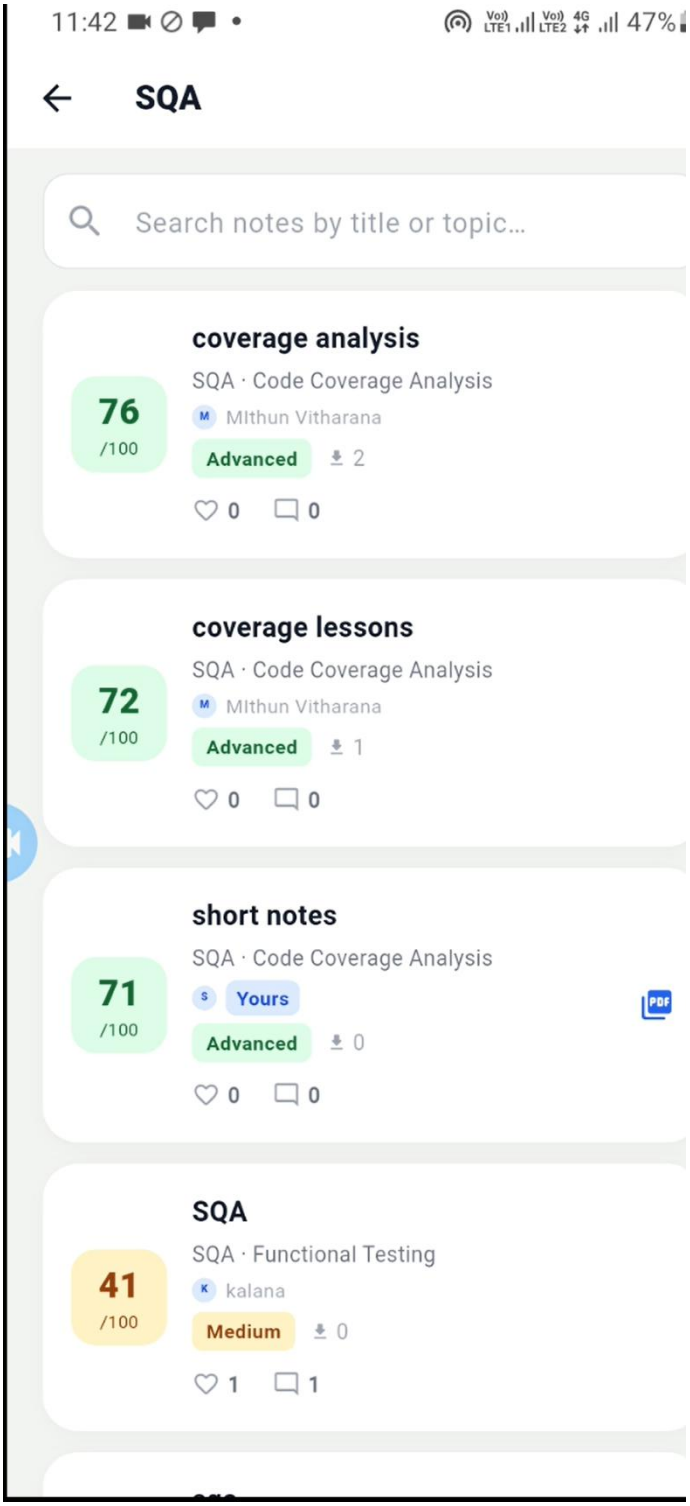
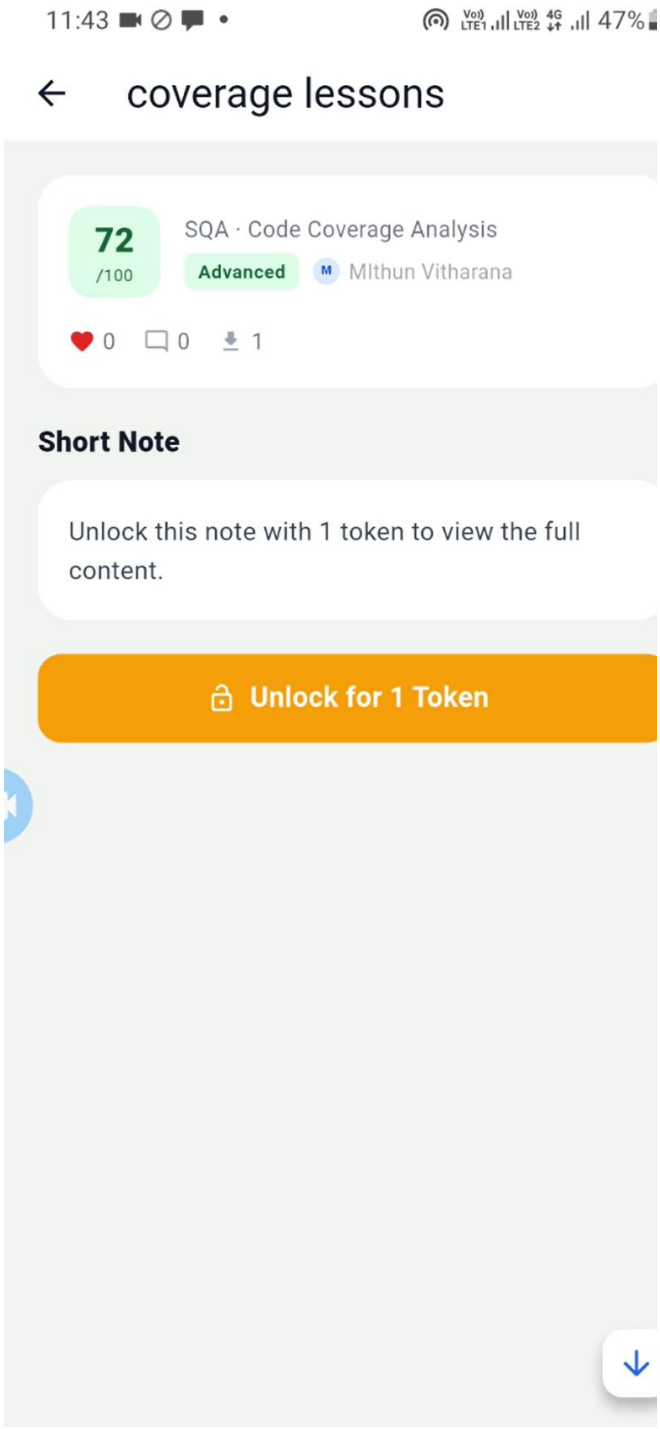


Figure A.4: Token-Based Academic Note Access



Appendix B – Academic Note Quality Assessment Rubric

The following rubric outlines the criteria proposed for evaluating the quality of shared academic notes.

Assessment Criterion	**Low (1)**	**Medium (3)**	**High (5)**
Conceptual Consistency	Unsupported content	Partially aligned	Strongly aligned with reference
Completeness	Major topics missing	Partial coverage	Essential topics covered
Clarity and Structure	Difficult to follow	Acceptably organised	Clear and logically organised
Reference Validity	No reliable references	Some supporting references	Appropriate and traceable references
Readability	Unclear language	Moderately understandable	Clear and suitable for learning level